

Appendix 2

Detail of monitoring

1. AIR MONITORING

(a) Particulars of person in charge of air monitoring (attach curriculum vitae)

(i) Name: _____

(ii) Qualifications: _____

(ii) Experience in conducting air monitoring _____

(b) Air Contaminant to be monitored:

No	Name of Contaminant	Method of Sampling	Method of Sampling
1	Hydrogen Sulphide		
2	Carbon Monoxide		
3	Carbon Dioxide		
4	Oil Mist		
5	Other toxic impurities e.g. dust and fumes		

MAN AND MEDICAL LOCKS AND APPOINTED DOCTOR

1. LOCKS

(a) Details of Locks

Type of Lock	Number of Locks	Dimensions
Man-Lock(s)		mØ xLength
Medical-Lock(s)		mØ xLength

2. LOCK ATTENDANTS

(i) Details of Man-Lock Attendants

Name of the man-lock attendant	
Training background	
Experience in handling man-locks	

(ii) Details of Medical-Lock Attendants

Name of the medical-lock attendant	
Details of training	
Details of experience in handling man-locks	

3. MAXIMUM WORKING PRESSURE

Maximum working pressure for the following locks: -

- i) Man-lockkpa
- ii) Medical-lockkpa
- iii) Working chamberkpa

4. APPOINTMENT MEDICAL PRACTITIONER(S):

- (a) Name: _____
- (b) Address of Practice: _____
- (c) Qualifications: _____

- (d) Training and experience in compressed air illness problems:

Appendix 3

Permissible exposure limit

SUBSTANCE (GAS, VAPOUR, FUME AND MIST)	PPM ^b	Mg/m ^{3c}
Acrylamide	-	0.3
Ammonia	25	17
Asphalt fumes	-	5
Benzene	10	32
Butane	800	1900
Calcium hydroxide	-	5
Calcium oxide	-	2
Carbon dioxide	5000	9000
Carbon monoxide	25	29
Carbon tetrachloride	5	31
Chloride	0.5	1.5
Coal tar pitch	-	0.2
Gasoline	300	890
Hydrogen cyanide	4.7	5 ^d
Hydrogen sulphide	10	14
Lead	-	0.15
LPG (Liquified Petroleum Gas	1000	1800
Nitric oxide	25	31
Nitrogen dioxide	3	506
Nitro-glycerine	0.05	0.46
Ozone	0.1	0.2
Phosgene	0.1	0.4
Sodium hydroxide	-	2
Sulphur dioxide	2	5.2
Sulphuric acid	-	1
Trinitrotoluene (TNT)	-	0.5
Welding fumes	-	5

Zinc chloride fume		
Zinc oxide fume		
DUST (OTHER THAN MINERAL DUST)		
Asbestos Amosite		1 fibre/ml
Chrysotile or other forms		1 fibre/ml
Fluorides		2.5 mg/m ³
MINERAL DUST		
Cement		
Nuisance particulates (e.g. calcium carbonate gypsum)		10 mg/m ³
Total dust		10 mg/m ³
Respirable dust		10 mg/m ³

Note:

- Permissible Exposure Limits refer to the time weighted average concentrations of airborne chemical substance for an 8-hour workday, to which workers can be repeatedly exposed without adverse health effects.
- Parts of vapour or gas per million parts of contaminated air by volume at 25°C and 760 mm Hg pressure.
- Approximate milligrams of substance per cubic metre of air.
- Capital letter C denotes ceiling limit which is the concentration that should not be exceeded even instantaneously.

Appendix 4

Quality Of Compressed Air In Working Chamber Man-Lock And Medical Lock

Carbon monoxide	not more than 10 parts per million
Carbon dioxide	not more than 500 parts per million
Oil	not more than 1 mg/m ³
Odour and cleanliness odour and not	As far as is practicable, the air should be free from all contamination by dust, dirt, and metallic particles and should contain any other toxic substance



Appendix 5

Lock Attendant's Register

Date: _____

Dry Bulb Temperature:

Wet Bulb Temperature:

Construction Site: _____

Name of the Lock Attendant: _____

Name of Worker	NRIC/Work Permit Number	Date & Time of Last Decompression	Compression		Maximum Working Pressure	Time in Working Chamber		Decompression		
			Times of entry into working chamber	Working Pressure		Hour	Min	Time decompression commenced	Time decompression finished	Decompression time

The time recorded on this Form must be taken from the clock provided for the use of the man-lock attendant.

Record all times as a.m or p.m

Appendix 6

Compression Procedure

1. Every employee going under air pressure for the first time should be instructed on the precautions to take during compression.
2. During the compression of employees, the pressure should not be increased to more than 0.2 bar within the first minute. The pressure should be held at 0.2 bar and again at 0.5 bar sufficiently long to determine if any employees are experiencing discomfort.
3. After the first minute the pressure should be raised uniformly and at a rate not to exceed 0.7 bar per minute.
4. If any employee complains of discomfort, the pressure should be held to determine if the symptoms are relieved. If, after 5 minutes the discomfort does not disappear, the man-lock attendant should gradually reduce the pressure until the employee signals that the discomfort has ceased. If he does not indicate that the discomfort has disappeared, the man-lock attendant should reduce the pressure to atmospheric and the employee should be released from the lock and required to report to the medical lock attendant.



Appendix 7

Compressed Air Worker's Transfer Record

Name of the Lock Attendant: _____

No: _____

Date	Lock No. or		Pressur e							Total decompression time in minutes	Lock Attendant's Signature
	When Entering	When leaving (if different)		Hour	Mins	Hours	Mins	Hours	Mins		



This record is to be retained by the person to whom it is issued. Entries are to be made by the man-lock attendant who shall also make the necessary entries in the Lock Attendant's Register (Appendix 4)

Appendix 8

Compressed Air Worker's Medical Record

This Form should be completed by the appointed medical practitioner in respect of a compressed air worker within 8 weeks after starting to be employed in compressed air work.

Name of Worker:		Date of Birth:		Sex:
Residential Address: _____				
Name & Address of Employer: _____				
Contact No: _____				
Date Employed: _____				
Previous Compressed Air Experience:				
Employer	Dates (From/To)	Maximum Working Pressure	Any Compressed Air illness (Type I/II)	Radiographic examination results and dates

(1) MEDICAL HISTORY

Ever declared unfit for employment in compressed air? YES / NO

Currently on any drugs or medication? YES / NO

Operations: _____

Injuries: _____

Bronchitis	Yes/No	Asthma	Yes/No	Hay fever	Yes/No
Sinusitis	Yes/No	Ear trouble	Yes/No	Tuberculosis	Yes/No
Hernia	Yes/No	Fits	Yes/No	Psychiatric disorders	Yes/No
Heart trouble	Yes/No	Hypertension	Yes/No	Diabetes	Yes/No
Peptic ulcer problem	Yes/No	Bone/joint disease	Yes/No		

(2) MEDICAL EXAMINATION

Height (Standing without shoes): _____

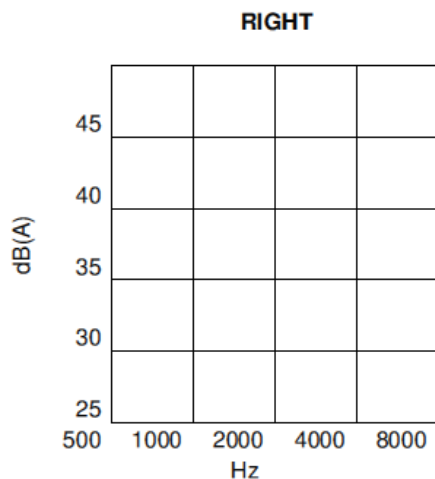
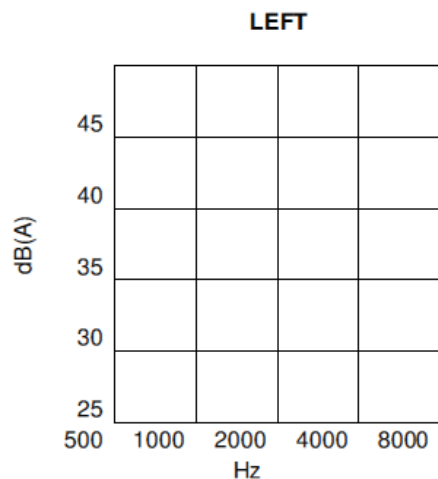
Weight (In light clothing without using beam or lever scale): _____

Body Fat: _____ %

Heart: _____ Pulse: _____ Blood Pressure: _____

Lungs: _____ Joints: _____

Ears: _____ Nose: _____ Sinuses: _____



Abdomen: _____

Nervous System: _____

Hb: _____

PBF: _____

ECG (If indicated): _____

Sickle cell trait (If indicated): _____

(3) RADIOGRAPHIC EXAMINATION

Chest X-ray: _____

Date of Examination: _____

Normal / Abnormal *

If Abnormal, attach a copy of report

Skeletal X-ray: (Including shoulder, hip

Date of Examination: _____

and knee joints)

Normal / Abnormal *

If Abnormal, attach a copy of report

(4) TEST IN LOCK:

(5) Fit for Employment in compressed air : YES/NO

REMARKS:

Doctor's name & Signature

(6) ADD ANY FURTHER DETAILS OR EXAMINATION RESULTS WITH DATES:

Date

Examination (given details)

Remarks

* Delete if not applicable.

Appendix 9

Compressed air illness notification form

Name of contractor: _____ Date _____ Employed: _____

Full Name of the Worker: _____ No: _____

Occupation: _____ Activity: _____
Manual/Supervisory/Sedentary

Maximum _____ Working _____ Pressure: _____

If _____ multiple _____ exposure, _____ how _____ many _____ ?

Total _____ exposure _____ time: _____

Was decompressed according to approved tablets ? YES/NO

New Starter: YES/NO

“Cold” Infection: Head or Chest YES/NO

Onset of symptoms after decompression: _____ Minutes/hour
(Date: _____)

TYPE I (BENDS)

Site _____ of _____ pain: _____

Any signs of symptoms other than pain (skin motting or irrigation): 1

TYPE II (If Yes, tick in the appropriate box for details)

Neurosurgical:

-
- | | | |
|---|-------------------------------------|--------------------------------------|
| ☐ Unconsciousness | ☐ Drowsiness | ☐ Headache |
| ☐ Vertigo | ☐ Tinnitus | ☐ Convulsions |
| ☐ Visual disorders | | |
| ☐ Paralysis (Indicate site or level) : _____ | | |
| ☐ Loss of sensation (Indicate site or level) : _____ | | |

Appendix 10

Compressed Air Workers Individual Record

Part I : General

This form should be completed by the appointed medical practitioner in respect of a compressed air worker.

Period under consideration: _____ to

Name of the worker: _____ NRIC/Work Permit No:

Name and Address of the contractor:

Date employed: _____ Date left employment: _____

(If applicable)

Shift Worker: YES/NO *

Part II : Summary

Compressed Air Work Record during month considered:

Maximum Working Pressure:

Total number of decompressions from:

Number of direct (conventional) decompression from:

(a) WP < 1.9 bars : _____

(b) WP > 1.9 bars : _____

* Delete if not applicable

WP : Working Pressure

☐ Bladder deficit

☐ Bowel deficit

Cardio-Respiratory:

Yes / No

☐ Substantial discomfort/Chest pain

☐ Breathlessness

☐ CO

Abdominal:

Yes / No

☐ Pain (Indicate site) : _____

☐ Diarrhoea

Systemic:

Yes / No

☐ Fatigue / Malaise

☐ Fever

☐ Nausea / vomit

☐ Other symptoms : _____

TREATMENT

Time Commenced: _____ Time Completed:



Pressure or Relief: _____ Highest Pressure used:

Method of Decompression: _____ Number of Decompressions:

Residual symptoms:

Fit for employment in compressed air again: YES/NO

Evidence of recurrence after leaving medical centre: YES/NO

Signature of the notifying Doctor

For decompressions resulting in compressed air illness symptoms, give the following details
:

Date	Maximum Working Pressure	Symptoms Type I / Type II	Number of Therapeutic Compressions Required

Type III : Detailed Records



Name of the Worker :

Permit : _____

Date	Length of Shift	Maximum Working Pressure	Compressed Air Illness		Symptom Not Treated
			Type I	Type II	

Appendix 11

Pressure and conversion

1. All pressure quoted in this guidebook are gauge pressure.

2. The conversion factors are:

1 bar	=	105 N/m ²	1 bar	=	14.5 psi
1 m	=	3.281 ft	1 ft	=	0.3048 m
1 mm	=	0.03937	1 in	=	25.4 mm

Appendix 12

Monthly Tunnel Checklist

1) Pre – Tunnelling Works:

Sl. No	Item	Compliance Status	Reference Code BS 6164 / Clause	Remarks	Photo Evidence
1	Whether the following method statements have been approved by the Employer / Engineer and the comments / control measures required are in place and duly verified for Initial drive Main drive Cutter head intervention Instrumentation and monitoring Compensatory grouting / secondary grouting Post tunnelling defects like cracks, damage, leakages etc.				
2.	Whether designer confirmation for the extent of TBM advance for the initial drive has been taken and the record available for verification.				
3.	Whether the facility provided for first aid and ambulance is acceptable to Employer / Engineer.				
4.	Do necessary measures have been taken to prevent shaft against flooding and adequate pumping facility is made available in the shaft.				
5.	The action plan for emergency evacuation and rescue operation is satisfactory and the flow chart with contact numbers for emergency has been submitted and acceptable to Employer / Engineer.				
6.	Proposal for firefighting has been submitted and acceptable to Employer / Engineer.				

7.	Provision is made for firefighting equipment is sufficient.				
8.	Emergency lighting is provided along the tunnel with battery backup.				
9.	Does the secondary power back up for TBM are available and adequate to runs the essential circuit in case of accidental failure of main power supply.				
10.	Whether the ground treatment at break-in and break-out portions has been completed.				
11.	Whether existing building survey along the Tunnel alignment has been completed.				
12.	Does any form of structural support for the buildings and structures in the zone of influence are required along the alignment.				
13.	Whether any supporting of buildings as per damage assessment has been carried out at site.				
14.	Whether all the study and survey for locating underground services which may affect the tunnel has been undertaken and found satisfactory.				
15.	Whether all the bore wells within the influence zone has been identified and plugged to the satisfaction of Employer / Engineer.				
16.	Whether sufficient numbers of boreholes for GIR have been taken along the alignment and for sufficient depth.				
17.	Whether rainfall data of previous years is available for guidance.				
18.	Whether ground water has been tested for salinity and other harmful chemicals and gases.				
19.	Whether the facility provided for cold water supply in the shaft is adequate for the workers.				

20.	Whether face pressure calculations have been submitted and acceptable to Employer / Engineer.				
21.	Has a gas monitoring system has been established by the following systems? a) fixed system b) portable system				
22.	Records for calibration of gas detection equipment are available.				
23.	Is entry into the tunnel controlled by tally board and which prevents non-inducted person entering the tunnel.				
24.	Has pre-employment health screening for workers has been completed and list submitted to Employer / Engineer.				
25.	Is there evidence of the competency of supervisors and operators, verified and found acceptable by Employer / Engineer.				
26.	Is the competency of electrician verified and acceptable to Employer / Engineer.				
27.	Whether the organisation chart for the work has been submitted and persons are interviewed for their basic knowledge and communication skills.				
28.	Has an assessment of the hazardous nature of chemicals and conditioners has been undertaken and found acceptable to Employer / Engineer.				
29.	Is the working visibility for the erector operator safe for working				
30.	Is the manlock is insulated against fire and the arrangement acceptable.				
31.	Has the testing of the compressed air installation been completed and acceptable to Employer / Engineer.				

32.	Whether following special hazards in compressed air working has been addressed and necessary actions are in place. a) Restriction of hot work. b) Usage of small cylinders for burning equipment.				
33.	Is there a permit to work system for welding and cutting which is acceptable to Employer / Engineer.				
34.	Does the hydraulic oil being used for TBM operation conform to low flammability category.				
35.	Are there sign boards indicting prohibition of petrol driven equipment displayed in the shaft / tunnel.				
36.	Is the network of cables properly arranged and is it possible to isolate them in case of fire so that most essential services can be run.				
37.	Is equipment provided for working hand held tools and lighting run on low voltage supplies. (110 volts)				
38.	Are the earthing arrangements satisfactory?				
39.	Have escape routes have been established and signed for clarity.				
40.	Whether third party testing of EPDM / hydrophilic gaskets have been completed and found satisfactory.				
41.	Whether approval for annulus grout material / operational grout mix plant is confirmed by Employer / Engineer.				
42.	Has the SAT has been completed and confirmed by Employer / Engineer.				
43.	Whether sufficient number of relevant segments of Type-1 & 2 which are more than 28 days old are available to start the work.				

44.	Have the facilities for the ventilation system been installed and commissioned to the satisfaction of Employer / Engineer				
45.	Are the necessary methods for control of Loco movement with signals and switches established?				
46.	Whether the tunnel guidance system has been commissioned.				
47.	Have the crew required for intervention has been Identified and verified that they possess sufficient training to effectively deal with intervention. Are the declared medically fit?				
48.	Whether the scheme for muck removal and disposal has been submitted and acceptable to Employer / Engineer				
49.	Whether the status of reaction frame and tunnel eye seal is satisfactory for operation.				
50.	Has access and egress to the shaft been safely erected and properly identified with alternative route for an emergency.				
51.	Have risk assessments and control measures been developed and are they adequate to manage the hazards identified.				
52.	Whether a agreed procedure for precautionary measures that need to be taken to keep the TBM face active during long idling of TBM to avoid the possible surface settlement have been documented and acceptable to Employer / Engineer.				
53.	Checks for LOCO Operation: a) Has braking test of the Loco is done.				

	b) Are automatic over speed brakes required and duly incorporated. c) Have warning lights been installed on the lead vehicle. d) Is there a format for daily functional brake check been established? e) are tracks are properly supported and secured against displacement.				
54.	Checks for Belt Conveyor: a) Is the belt conveyor system fully guarded. b) Have belt weighing equipment been fitted to the conveyors. c) Is the starting of the conveyor possible only from a single controlled position. d) Is the belt conveyor system fire resistant. e) are emergency stop buttons provided at suitable close intervals along accessible length. f) Is there provision of an audible start-up alarm installed and operational. g) Is there provision of anti-runback devices for inclined conveyer, installed and operational.				

2) During Tunnelling Works:

Sl. No	Item	Compliance Status	Reference Code BS 6164 / Clause	Remarks	Photo Evidence
55.	Provision for CCTV camera has been made.				
56.	Whether the shift details for staff working in the tunnel acceptable.				
57.	Has the means of site communications been addressed and acceptable to Employer / Engineer.				
58.	Is an alert method in case of Emergencies been put into place and acceptable to Employer / Engineer				
59.	Has provision for Whole Body protection been made for hot work.				
60.	Are there sufficient number of Self Rescuer available which should last for at least 60 minutes?				
61.	Are there sufficient first aid facilities including stretchers available within working area.				
62.	Is there Provision for dedicated skip, train or vehicle available for transporting injured persons.				
63.	Are there arrangements to provide adequate lighting levels, is it acceptable to Employer / Engineer				
64.	Whether the facility for excavation volume assessment is in place and acceptable to Employer / Engineer				
65.	Whether the means for containment in case of settlement has been addressed and found acceptable to Employer / Engineer				
66.	Whether the methods for compensation / secondary grouting is submitted and acceptable to Employer / Engineer				
67.	Whether required procedure for clearing Inundation of Tunnel has been addressed and acceptable to Employer / Engineer				
68.	Is the Storage of Combustible materials well demarcated with display board.				
69.	Is there provision for Smoke control in the Tunnel.				
70.	Whether onsite procedures for renovation and repair are established and acceptable to Employer / Engineer.				

APPENDIX 13

	CHENNAI METRO RAIL LIMITED
CONSTRUCTION / INSTALLATION PHASE OCCUPATIONAL HEALTH & SAFETY TUNNEL PLAN	
Contractor Name	
Project Name	
1	Application of this document
2	Purpose of this document
3	Scope
4	Referance Publications
5	Terms and definitions
6	OH&S Management System Requirements
7	General Requirements
8	Contractor OH&S policy
9	Drug & Alcohol Policy Statement
10	Planning and Implementation Planning and Implementation Interface Management Management of Change Procurement Sub-contractor Selection Construction Management Process Overview
11	OH&S Objectives and programme(s)
12	Legal and other requirement Legal and other requirement International Standards, Guidelines & ISO Certifications
13	Implementation and operation Contractor's OHS&E Organization Resources, roles, responsibility, accountability and authority Roles & Responsibilities General duties of the contractor and others General duties of persons employed Contractors and sub-contractors responsibilities Duties of the OH&S Persons Powers of OH&S persons Competence, training and awareness OHS&E personnel NONO from the Employer/Engineer
14	Method Statement
15	Lift Plan

16	Hazard identification, risk assessment and determining controls Hazard identification, risk assessment and determining controls Hazard Log & Risk Register Risk Assessment COSHH Assessment Design Risk Assessment Risk assessment production and submittal
17	Designer's Role
18	ID card and First day at work, OH&S Orientation training
19	Safety Training and Safety Promotion Occupational health and safety training Induction training Refresher Training Specific Training Tool box talks Occupational Safety Promotion
20	OH&S communication Occupational Health and safety signage Communication, Participation and consultation Contractor's site OHS&E committees Sub contractor awards
21	Documentation Management system procedure Management System Management System procedures Corporate Operating Procedures Standard Operating Procedures Standard Operating Procedures Safety Standard Operating Procedures Environmental Standard Forms Safety Standard Forms Environmental Control of documents Operational control Operational procedures Document Control Control of Records Filing System
22	Performance measurement and monitoring Performance measurement and monitoring Surveillance Exercise's Site Safety Inspections Plant & Equipment Inspection
23	Request for Inspection Evaluation of compliance
24	Reporting of accidents and Dangerous Occurrence Contractor responsibility Reportable accidents Dangerous occurrences Reporting of fire by the contractor

	Reporting to Employer/Engineer
25	Accident investigation General Recommended actions in incident investigation
26	Accident Statistics Collection of accident statistics Calculation of man-days lost –construction accident statistics Calculation of accident frequency rate (AFR)
27	Nonconformity, corrective action and preventive action Control of records
28	OH&S Audit MARS Audit ISO Certification Audit Engineer Audit External Quarterly OH&S Audit Period of Quarterly Audit Areas of competence of audit team
29	Management review
30	Industrial Health and Welfare Physical Fitness of Workforce Medical Facilities Medical Examination Occupational Health Centre Medical Staff HIV/AIDS prevention control Prevention of Mosquito breeding Alcohol and drugs Noise Ventilation Radiation
31	Welfare measures for workers Latrine and urinal accommodation Moving sites Canteen Drinking water Labour Accommodation • Living Space • Toilet/Washing Facilities • Food Preparation/Cooking Area • Lighting and ventilation • Evacuation • Evacuation procedure • First Aid/Emergency • Hazardous area • Fire safety and fire fighting • Mechanical and electrical safety • House keeping • Transportation • Pest Control

	• Camp/ Accommodation Supervisor
32	Working at height General Use of scaffolds Use of ladders Safety harnesses/Fall Arresters
33	Excavations General requirements on excavation work Duties of engineer on excavation Access and egress from excavation Undermining nearby structures Avoiding underground services
34	Lifting Operations Lifting appliances Crane safety Equipment Lifting gear
35	Site Electricity General Design considerations Cables Maintenance Lighting Illumination Electrical power circuits Installation of electric wiring and power lines Residual current circuit breakers and overcurrent protective devices Prohibition on use of fuse Industrial plug and socket outlet Distribution board and socket out assembly Electrical installations and equipment used underground or in confined space. Monthly Electrical Safety Audit
36	Welding, gouging and cutting Oxy-fuel gas processes Arc Cutting, gouging and welding processes Compressed gases storage Handling of compressed gas cylinders Regulators Hoses Training and instruction
37	Machinery Machinery Fencing Maintenance Air receivers Woodworking machines Abrasive wheels
38	Plant and equipment operations Test and periodical examination Automatic safe load indicators
39	Personal Protective Equipment Head protection Eye protection

	Whole body protection Respiratory protection High –visibility clothing Hearing protection Self-Rescuers
40	First Aid provisions and procedures First aid bases First Aid boxes First aid facilities First Aid rooms First aid Training Stretchers Ambulance van
41	Blasting Operation Authorisation for blasting Risk assessment and method statements
42	Demolition General duty of the contractor Preparation of demolition Protection of adjacent structures Removal of load bearing structures Demolition of walls, partitions, etc. Access to floor Barricades, catch platforms and warning signs Mechanical method of demolition
43	False work /Formwork General duty of the contractor General Requirements Supports and shores Appointment of formwork supervisor Duties of formwork supervisor Concrete work Dismantling Steel Reinforcement Reshoring
44	Piling and Diaphragm walls Stability of adjacent structures Inspection Pile driver not in use Pile testing Footing
45	Work adjacent to live Railways
46	Work adjacent to live roadways
47	Traffic Management Legal permission Regulatory signs Warning signs Delineators Traffic Cones and cylinders

	Drums Barricades Yow away vehicle Cleaning of roads Spoil Removal
48	House Keeping Accumulation of debris Method of removal of debris, etc. Floor opening for removal of debris Debris collection area
49	Over Head protection
50	Slipping, Tripping, Cutting, Drowning and Falling Hazards
51	Manual Lifting
52	Hand Tools and Power Tools
53	Dangerous and Harmful Environment
54	Corrosive substances
55	Batching Plant and Casting Yard
56	Radiation
57	Smoking
58	Mobile Phones, Radios and Audio Equipment's
59	Dust ,Gases, fumes
60	Vehicular Traffic
61	Explosives General duty of the contractor Handling of explosives Smoking, open lights etc. Opening Packages Deepening holes Size of holes Removing cartridge wrappers Loading and other operations Loading and tamping Work of firing of any blast Warning and retreat Return to blast area Misfires Operations during Thunderstorm
62	Floor Openings
63	Evaluation , selection and control of subcontractors
64	CCTV

65	Site Security Hoardings Personnel access Vehicular access Site patrols Security Guards
66	Flooding Flooding Evacuation and Rescue procedure
67	Work permit system Implementation of permit to work No high risk construction work without permit to work Application for permit to work Evaluation of permit to work Issue of permit to work Posting of permit to work and supervisor's duty Monitoring of work Duty to report incompatible work Revocation of permit to work
68	Grievance and Disciplinary
69	Heavy Plant operations
70	Site Perimeter Hoarding
71	Emergency Preparedness and Response
72	Warning Notice Access /Egress Plan & Control of personnel, Staff & Workmen, Check in /Check out
73	Investigation and information gathering
74	Excavation and ground support
75	Bored Tunnels
76	Gassy Operations
77	Tunnel plant Erector Safety , Cutter Head safety
78	Instrumentation Monitoring
79	Safety requirements in all stages of construction
80	Ground support
81	Permanent Support
82	Dealing with Water
83	Tunnelling boring machines and road headers
84	Inundation Precautions Escape
85	Methane General Occurrence Explosion Characteristics Detection and monitoring Danger Levels

	Sources of ignition Explosion Protection
86	Diesel Engines
87	Hydraulic Plant
88	Shaft and Hoist
89	Rail Haulage
90	Flooding
91	Illumination
92	Ventilation General Control of contamination Oxygen Deficiency Explosive atmospheres
93	Under Ground communication
94	Noise and Vibration Tunnel face Access routes Noise sources Noise reduction Communications Vibration Level of vibration Whole body vibration exposure Level of hand arm vibration exposure
95	Transport and loading Rail Haulage Clearance Propelling Locomotives Fuelling Battery locomotives Potentially explosive atmosphere Rolling stock Couplings and buffers Track Winch haulage Rubber Tyred vehicles Belt conveyors Slurry pumping Mixing system Tipping and disposal
96	Tunnel Plant Pneumatic Plant Hydraulically operated plant Internal combustion engines Diesel engines Siting of engines Concerting plant Water drainage pumps

97	Fire Prevention, protection, fighting system and Rescue Fire Mains and hose connections Fire extinguishing system Routine testing and maintenance of fire protection equipment Vulnerable items and locations Escape routes Fire Fighting's and rescue Fire services and operational capacity Fire Fighting equipment Emergency control facilities Bridge heads Raising the alarm Alarm procedure Site training Lighting Smoke control Rescue facilities Self Rescuers Accounting personnel Storage of flammable liquids
98	Access Walkways Access maintenance Gradients
	Drilling
99	Compressed Air Environment General duty of contractor Safe system of work Warning Notice Use of bulkhead Plant and equipment Supply of air Power Source Temperature and humidity Means of communications Appointment of competent person for compressed air works Identification badges Conditions for persons working in compressed air environment Condition for multiple entries in compressed air environment Prohibition on consumption of alcohol and smoking Man-Locks Compression and decompression procedure and use of man lock Medical Locks Attendance at man lock Attendance at medical lock Appointment of medical practitioner Resting facilities and First Aid Room Maintenance of records Training of personnel

100	Monitoring of the tunnel atmosphere
101	Tunnel Track Maintenance Roles and competencies Tunnel track work permit Tunnel track inspection and maintenance
102	Pathways
103	Permissible exposure Limits
104	Dust
105	Protection against dust
106	Gaseous contamination
107	Emission from power units
108	Heat
109	Visitor's to site
110	OH&S Submittals to the Employer/Engineer
111	Appendix Emergency Preparedness Flow Chart, Emergency Evacuation Plan layout with Emergency Contact Number General instruction for Entry into tunnel including visitor induction & Tunnel Visitor pass Tally board

EVALUATION AND QUALIFICATION CRITERIA

I. Eligibility/Qualification Criteria

The Bidders who have the following Eligibility criteria will only be considered for Technical Bid Evaluation:

S. No	Criteria	Specific Requirement	Documents Required
1	Earnest Money Deposit (EMD)	Bidders must submit Earnest Money Deposit (EMD) of INR 94,000/- (Rupees Ninety-four thousand only) in form of Demand Draft / NEFT / RTGS in favor of Chennai Metro Rail Limited payable at Chennai.	Bidder must upload scanned copy of EMD along with bid and has to ensure delivery of hardcopy to the Buyer <u>within 2 days</u> of Bid End date / Bid Opening date. The bidder seeking EMD exemption, must submit the valid supporting document as per the provisions of GeM terms.
2	Power of Attorney	Board resolution or power of attorney in the name of an Authorized Signatory of the Bidder stating that he is authorized to execute documents and to undertake any activity associated with the Bidder's Proposal.	Notarized copy of Board Resolution or Power of Attorney in the name of the Authorized signatory.
3	Legal Entity	The following Indian Firms are allowed to participate in the bid process: I. Companies registered under Companies Act 1956 or 2013 II. Partnership firms registered under Limited Liability Partnerships (registered under LLP Act, 2008) III. Partnership firms registered under Indian Partnership Act, 1932 IV. Sole Proprietorship firm. Note: Consortiums or Joint Ventures are not allowed participate.	Documents as applicable to be submitted along with Annexure-A • Certificate of Incorporation. • Memorandum of Association. • Articles of Association. • Partnership Deed • Corporate Identity Number Certificate.
4	Statutory Tax Registrations	The Bidder should have: I. Valid PAN Number II. Valid GST Number III. Valid EPFO registration number IV. Valid ESIC registration number	Copy of PAN Card Copy of GST Registration Certificate Copy of EPFO registration certificate Copy of ESIC registration certificate
5	Average Annual Turnover from General Consultancy / PMC or DDC	The Bidder must have an average annual turnover in the last three financial years (F.Y. 2022-23, 2023-24 and 2024-25 respectively) from consultancy services of not less than INR 24 Lakhs.	Certificate from a Chartered Accountant (CA) along with audited financial statements which includes profit and loss account and balance

Consultancy Assignment for Conducting an Independent Safety Audit to Enhance Compliance with Safe Systems of Work in Chennai Metro Rail Limited (CMRL) Phase-2 Civil and Track Projects – Section A

	or both or safety or Quality audit related Services		sheet for the last three financial years to be attached as Annexure-B.
6	Net worth Requirements	The bidder should have a positive Net worth during the last audited financial year (ending with March'25).	
7	**Similar Assignment	The bidder must have successfully executed at least two similar assignments in the field of infrastructure projects within the last seven (07) years. These assignments should preferably include projects such as metros (elevated or underground), railways, airports, bridges, or road construction. For the purpose of this assignment, which is rare in the market, **similar assignment is defined as follows, "Any agency who executed General Consultancy / PMC or DDC or both or safety or quality audit in compliance to ISO 45001 or ISO 9001 any further amendments.	Work order and completion certificates from the client; In case of Ongoing projects, Work order and a letter from the client about the satisfactory compliance till date may be provided.
8	Restriction on Procurement due to National Security	The bidder should not be from a country which shares Land Border (Restriction on Procurement due to National Security) with India.	An undertaking to this effect should be submitted by the bidder on its letter head. To be attached as Annexure-D
9	Sub-Contract	The bidder should bid individually and not as a Joint Venture/Consortium. Bidder should not subcontract the consultancy assignment.	Self-declaration duly signed by the authorized signatory.
10	Integrity Pact	Integrity Pact shall have to be complied by the bidders	An undertaking to this effect should be submitted by the bidder on its letter head. To be attached as Annexure-E
11	Blacklisting and Debarment	The Bidder shall not be under a declaration of ineligibility / banned / blacklisted by any Government / Government Undertaking / Public Sector Organization / Statutory Body / any organization where Government is the majority stake	Self-declaration by the bidder on its letterhead duly signed, sealed/ stamped. To be attached as Annexure-C

		holder as on the last date of submission of the Bid.	
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The bidder not qualifying the above mentioned minimum qualification criteria will not be considered for further process of technical bid evaluation, and such bids shall be treated as nonresponsive.

II. Evaluation of bids.

- Only those bidders who qualify in technical evaluation and satisfy all the 'Qualifying Requirements', would be considered eligible for opening of financial bids.
- A Consulting firm will be selected through 'Least Cost Selection' (LCS). The minimum Technical Score required to be achieved is 70 Points in the following technical evaluation criteria.

A. Technical Evaluation Criteria and Scoring

The Technical Proposal will be evaluated based on the following criteria. Each of the parameters is detailed in the subsequent sections. Bidders must score at least 70% (i.e. 70 points) in the sum of Evaluation Parameter (i.e Sl.No. 1, 2 & 3), to qualify for the financial bid evaluation.

Sl. No.	Evaluation parameter	Total Marks
1	Firm Credentials	30 points
2	Team Composition and Expertise	50 points
3	Quality- Approach & Methodology and Presentation	20 points
	Total	100 points

Sl. No.	Description	Maximum Points	Submission Requirements
1.	Firm Credentials	30 points	
(i)	Experience of the Bidder: Total years of experience in providing consultancy services related to General Consultancy/Detailed Design Consultancy (DDC)/ Safety Audits. 4 years: 10 points 5-7 years: additional 2.5 points More than 7 years: additional 2.5 points	15 points	Form TECH-2
(ii)	Similar Experience: Experience in **Similar Assignments in the field of infrastructure projects preferably projects such as metros (elevated or underground), railways, airports, bridges, or road construction in the last Seven (07) years. 2 Assignments : 10 points 3-5 Assignments : additional 2.5 points More than 5 Assignments : additional 2.5 points	15 points	
2.	Team Composition and Expertise (Key Personnels)	50 points	Form TECH-3

Consultancy Assignment for Conducting an Independent Safety Audit to Enhance Compliance with Safe Systems of Work in Chennai Metro Rail Limited (CMRL) Phase-2 Civil and Track Projects – Section A

	It is desired to have the following senior key professionals, based on the years of experience with supporting team members, marks will be awarded based on relevant National and International experience, domain knowledge, educational qualifications, association with firm, etc.													
	<table><tr><td colspan="2">Key Personnel 1 – Lead Auditor</td></tr><tr><td>(a) Educational Qualification</td><td rowspan="2">7.5 points</td></tr><tr><td>BE/BTech/ME/MTech in civil or mining or structural or design. Preferably Civil department -</td></tr><tr><td>(b) Professional Qualification Certified Lead Auditor ISO 9001, 45001, etc.</td><td>2.5 points</td></tr><tr><td>(c) Overall Professional Experience 15 years: 3 points 16 -18 years: additional 1 point Above 18 years: additional 1 point</td><td>5 points</td></tr><tr><td>(d) Experience in relevant works (refer TOR clause 10) 08 years: 6 points 9 -10 years: additional 2 points Above 10 years: additional 2points</td><td>10 points</td></tr></table>	Key Personnel 1 – Lead Auditor		(a) Educational Qualification	7.5 points	BE/BTech/ME/MTech in civil or mining or structural or design. Preferably Civil department -	(b) Professional Qualification Certified Lead Auditor ISO 9001, 45001, etc.	2.5 points	(c) Overall Professional Experience 15 years: 3 points 16 -18 years: additional 1 point Above 18 years: additional 1 point	5 points	(d) Experience in relevant works (refer TOR clause 10) 08 years: 6 points 9 -10 years: additional 2 points Above 10 years: additional 2points	10 points	25 points	
Key Personnel 1 – Lead Auditor														
(a) Educational Qualification	7.5 points													
BE/BTech/ME/MTech in civil or mining or structural or design. Preferably Civil department -														
(b) Professional Qualification Certified Lead Auditor ISO 9001, 45001, etc.	2.5 points													
(c) Overall Professional Experience 15 years: 3 points 16 -18 years: additional 1 point Above 18 years: additional 1 point	5 points													
(d) Experience in relevant works (refer TOR clause 10) 08 years: 6 points 9 -10 years: additional 2 points Above 10 years: additional 2points	10 points													
	<table><tr><td colspan="2">Key Personnel 2 – Deputy – Lead Auditor</td></tr><tr><td>(a) Educational Qualification BE/BTech/ME/MTech in Civil or Mechanical or Mining or Structural or Design. Preferably Civil department along with M.E / MTech in industrial safety engineering / RLI - PGDIS/ PG diploma in industrial safety approved by state technical board.</td><td>7.5 points</td></tr><tr><td>(b) Professional Qualification Certified Lead Auditor ISO 9001, 45001 or CSP or any other equivalent safety certification acceptable to the Employer.</td><td>2.5 points</td></tr><tr><td>(c) Overall Professional Experience 12 years: 3.0 points 12-14 years: additional 1.0 point Above 14 years: additional 1.0 point</td><td>5.0 points</td></tr><tr><td>(d) Experience in relevant works (refer TOR clause 10) – 6 years: 6 points 6 - 8 years: additional 2 points Above 8 years: additional 2 points</td><td>10 points</td></tr></table>	Key Personnel 2 – Deputy – Lead Auditor		(a) Educational Qualification BE/BTech/ME/MTech in Civil or Mechanical or Mining or Structural or Design. Preferably Civil department along with M.E / MTech in industrial safety engineering / RLI - PGDIS/ PG diploma in industrial safety approved by state technical board.	7.5 points	(b) Professional Qualification Certified Lead Auditor ISO 9001, 45001 or CSP or any other equivalent safety certification acceptable to the Employer.	2.5 points	(c) Overall Professional Experience 12 years: 3.0 points 12-14 years: additional 1.0 point Above 14 years: additional 1.0 point	5.0 points	(d) Experience in relevant works (refer TOR clause 10) – 6 years: 6 points 6 - 8 years: additional 2 points Above 8 years: additional 2 points	10 points	25 points		
Key Personnel 2 – Deputy – Lead Auditor														
(a) Educational Qualification BE/BTech/ME/MTech in Civil or Mechanical or Mining or Structural or Design. Preferably Civil department along with M.E / MTech in industrial safety engineering / RLI - PGDIS/ PG diploma in industrial safety approved by state technical board.	7.5 points													
(b) Professional Qualification Certified Lead Auditor ISO 9001, 45001 or CSP or any other equivalent safety certification acceptable to the Employer.	2.5 points													
(c) Overall Professional Experience 12 years: 3.0 points 12-14 years: additional 1.0 point Above 14 years: additional 1.0 point	5.0 points													
(d) Experience in relevant works (refer TOR clause 10) – 6 years: 6 points 6 - 8 years: additional 2 points Above 8 years: additional 2 points	10 points													

3	Quality- Approach & Methodology and Presentation Presentation regarding understanding of the project-objectives, Approach and Methodology, Work implementation schedule, <u>(PDF version copy of the presentation to be enclosed as Annexure)</u>	20 Points	Form TECH-4
(i)	Understanding of TOR, Approach & methodology to be adopted for the assignment	10 points	
(ii)	Work implementation and Staffing Schedule in accordance with the Terms of Reference of the assignment.	10 points	
	Total of Evaluation Parameter (Sl.No 1+ 2+3)	100 points	

Note:

- (i) **Similar Assignment: For the purpose of this assignment, which is rare in the market, similar assignment is defined as follows, “Any agency who executed General Consultancy / PMC or DDC or both or safety / Quality audit in compliance to ISO 45001 or ISO 9001 any further amendments.
- (ii) The bidder is required to submit Completion Certificates / Experience certificates from the Employer as supporting documents for each of the assignments exhibited. In case of Ongoing projects, Work order and a letter from the client about the satisfactory compliance may be provided.
- (iii) For the purpose of evaluation, the inflation of 5% every year (annual compounding) shall be applied for updating to the current price level i.e 28 days prior to the stipulated date of submission of the Bid as amended from time to time.
- (iv) The Inflation of 5% (annual compounding) is applicable to financial data and value of contracts.
- (v) The Proposed key personnel presented in the Bidder’s proposal may not be replaced without the prior approval of the Employer.
- (vi) The proposed key personnels’ age limit in the Consultant’s proposal shall fulfill the age limit criteria specified in TOR clause 10. “Qualification and Experience of Professional Staff”, if not fulfilled with the age limit criteria, the technical proposal shall be treated as non-responsive.

B. Evaluation of Financial bids:

The Financial bid shall be opened for only those bidders who have been found to be technically eligible. The least cost proposal (L-1) will be considered for award of contract. Financial bid should be checked by bidders to ensure conformance to the format provided in the bid document. If there is any discrepancy between words and figures in any part of the financial bid, the amount indicated in words will prevail.



Consultancy Assignment for Conducting an Independent Safety Audit to Enhance Compliance with Safe Systems of Work in Chennai Metro Rail Limited (CMRL) Phase-2 Civil and Track Projects – Section A

Form TECH-1: Technical Proposal Submission Form

Date : [insert date of Proposal submission]
GeM Bid No. : [insert Bid No.]
Project : [insert name of Project]
Assignment : [insert name of Assignment]

To: The Tender Inviting Authority,
Employer Address

Dear Sir/ Madam:

We, the undersigned, offer to provide the consulting Assignment for [insert name of assignment] in accordance with your Request for Proposals dated [insert GeM Bid No.] including addenda issued [Insert the number and issuing date of each addendum] and our Proposal.

We are hereby submitting our Proposal which includes [insert "this Technical Proposal, and a Financial Proposal sealed in a separate envelope"].

We hereby declare that:

- (a) All the information provided and statements made in this Proposal are true and we accept that any misinterpretation or misrepresentation contained in this Proposal may lead to our disqualification by the Employer.
- (b) Our Proposal shall be valid for 90 days from the date of Bid submission and remain binding upon us for the period of time specified.
- (c) We have no conflict of interest in accordance with clause
- (d) We understand that the Employer is not bound to accept any Proposal that the Employer receives.
- (e) We hereby certify that we have taken steps to ensure that no person acting for us or on our behalf will engage in any type of fraud and corruption.
- (f) We hereby certify and declare that if awarded with this assignment "no form of child labor or bonded labor shall be employed anywhere in the supply chains".
- (g) We hereby certify that the proposal submitted by me / us is strictly in accordance with the terms, conditions, Terms of Reference etc. as contained in the RFP document, and it is further certified that it does not contain any deviation to the aforesaid documents.

We remain,

Yours sincerely,

Authorized Signature : [insert signature of the person duly authorised to sign the Proposal, in full and initials]

Name and Title of Signatory : [insert full name and title of the person above]

Name of Consultant : [insert name of firm]

In the capacity of : [insert position of the person signing above]

Contact information : [insert mailing address]
[insert tel. no. with country and city codes]
[insert fax no. with country and city codes]
[insert email address]

[Person signing the Proposal shall have the Power of Attorney given by the Consultant to be included in the Technical Proposal.]

Form TECH-2: Consultant's Organization and Experience

Consultant's Organization

[Provide here a brief description of the organization (two pages) and experience in Enterprise-Wide Risk Management Consultancy assignment in the table below]

Sl. no	Name of Client	Scope of work	Period of contract (From/to)	Assignment awarded Value (in Rs.)	Contract Status		Proof of work	
					Continuing (Yes/No)	Date of Completion	Copy of LOA to be attached (Yes/No)	Completed work (Client certificate to be attached)
1.								
2.								
3.								
4.								

Note:

List of assignments mentioned above to be supported with Copy of purchase order / invoice / other supporting document along with completion certificate.

Date:

Signature.....

(Authorised signatory)

Place:

Full Name.....

Designation.....

Seal:

Consultancy Assignment for Conducting an Independent Safety Audit to Enhance Compliance with Safe Systems of Work in Chennai Metro Rail Limited (CMRL) Phase-2 Civil and Track Projects – Section A
Form TECH-3: Curriculum Vitae (CV) for Proposed Key Personnels

[The Consultant shall prepare the CV of each Key Personnel it proposes, using the format below for each Key Personnel].

1. General

Name of Key Personnel	[Insert full name]
Current Designation in the firm	
Proposed role in this assignment	
Date of Birth	[day/month/year]
Nationality	

2. Education

[list college/university or other specialized education, giving names of educational institutions, dates attended, degree(s)/diploma(s) obtained]

3. Employment record relevant to the assignment

Name of Company	Nature of consulting experience*	From Date	To Date	Brief work profile

*Nature of consulting experience should be as per Technical Evaluation Criteria

I, the undersigned, certify that to the best of my knowledge and belief, this CV correctly describes the qualifications, experience and other information of key personnel to be associated with the study. I understand that any willful misstatement described therein may lead to disqualification of the firm.

Date:

Signature.....

(Authorised signatory)

Place:

Full Name.....

Designation.....

Seal:

Form TECH-4: Description of Approach, Methodology and Work Plan for Performing the Assignment

[Technical approach, methodology and work plan are key components of the Technical Proposal. It is suggested that the Consultant presents its Technical Proposal divided into the following three chapters:

(a) Technical Approach and Methodology:

The Consultant shall explain its understanding of the objectives of the assignment, approach to the Assignment, methodology for carrying out the activities and obtaining the expected output, and the degree of detail of such output.

The Consultant shall highlight the problems being addressed and their importance, and explain the technical approach it would adopt to address them. The Consultant shall also explain the methodologies it proposes to adopt and highlight the compatibility of those methodologies with the proposed approach. Staffing for training shall also be explained, if so required in the TOR. A simple repeat/copy of the TOR will not be appropriate.

(b) Work Plan:

The Consultant shall propose the main activities of the assignment, their content and duration, phasing and interrelations, milestones (including interim approvals by the Employer), and delivery dates of the reports. The proposed work plan shall be consistent with the technical approach and methodology, showing understanding of the TOR and ability to translate them into a feasible working plan.

A list of the final documents, including reports, drawings, and tables to be delivered as final output, shall be included here.

(c) Organization and Staffing:

The Consultant shall propose the structure and composition of its team. The Consultant shall list the main disciplines of the assignment, the Key Personnels and Non-Key Personnels, and proposed technical and administrative support staff.



Consultancy Assignment for Conducting an Independent Safety Audit to Enhance Compliance with Safe Systems of Work in Chennai Metro Rail Limited (CMRL) Phase-2 Civil and Track Projects – Section A
Form FIN-1: Financial Proposal Submission Form

Date : [insert date of Proposal submission]
GeM BID No. : [insert BID No.]

To:

The Tender Inviting Authority,
Employer Address

Dear Sir/ Madam:

We, the undersigned, offer to provide the consulting Assignment for [insert name of assignment] for Chennai Metro Rail Limited in accordance with your Request for Proposals dated [insert Date] and its addendum.

We are hereby submitting our Financial Proposal for the sum of [insert amount(s) in words and figures].

We hereby declare that:

- (a) Our Financial Proposal shall be binding upon us subject to the modifications, if any, up to expiration of the validity period of the Proposal as mentioned in
- (b) We understand that the Employer is not bound to accept any Proposal that the Employer receives.

We remain,

Yours sincerely,

Authorized Signature : [insert signature of the person duly authorised to sign the Proposal, in full and initials]

Name and Title of Signatory : [insert full name and title of the person above]

Name of Consultant : [insert name of firm]

In the capacity of : [insert position of the person signing above]

Contact information : [insert mailing address]
[insert tel. no. with country and city codes]
[insert fax no. with country and city codes]
[insert email address]

Bidder Information Sheet
(To be submitted on Bidder's Letterhead)

Date : [insert date of Proposal submission]
 BID No. : [insert BID No.]

[Bidder shall provide the following information. The relevant documents shall be submitted as attachments hereto.]

Bidder's legal name: [insert full name]
Bidder's country of registration: [insert country of registration]
Bidder's year of incorporation: [insert year of incorporation]
Bidder's legal address in country of registration: [insert mailing address]
Bidder's authorized representative information Name: [insert full name] Address: [insert mailing address] Telephone/Fax numbers: [insert telephone/fax numbers, including country and city codes] E-mail address: [insert E-mail address]
Provide bank details of the bidder in the following format to enable us to return the EMD to unsuccessful bidder: a) Name of the Bank: b) Account Number: c) Kindly attach scanned copy of one Cheque book page

Date:

Signature.....
 (Authorised signatory)

Place:

Full Name.....
 Designation.....
 Seal:

Financial Statement
(To be submitted on Bidder's Letterhead)

Date : [insert date of Proposal submission]
 BID No. : [insert BID No.]

Type of Financial information	Historic Information for Previous Years in INR				
	2025-24	2024-23	2023-22	2022-21	2021-20
1. Total Assets (TA)					
2. Total Liabilities (TL)					
3. Net Worth (1-2) (NW)					
4. Current Assets (CA)					
5. Current Liabilities (CL)					
6. Working Capital (5-6) (WC)					
7. Annual Turnover in Consultancy services					
8. Profits Before Taxes (PBT)					
9. Profits After Taxes (PAT)					

Date:

Signature.....

(Authorised signatory)

Place:

Full Name.....

Designation.....

Seal:

Note: Bidder's shall furnish the above information duly certified by the Chartered Accountant / Statutory Auditor(s) along with audited financial statements for the financial years.

Declaration for No Blacklisting
(To be submitted on Bidder's Letterhead)

To
The Tender Inviting Authority,
Employer Address

Subject: Submission of No Black Listing Self-Declaration for Bid No."..."

Dear Sir

We do hereby declare and affirm that we have not been blacklisted by Central / any State Government / PSU's or any regulatory bodies as on the date of bid submission.

Date:	Signature..... (Authorised signatory)
Place:	Full Name..... Designation..... Seal:

Undertaking regarding restrictions on procurement from a bidder of a country which shares a land border with India
(To be submitted on Bidder's Letterhead)

To
The Tender Inviting Authority,
Employer Address

Subject: Submission of Undertaking regarding restrictions on procurement from a bidder of a country which shares a land border with India for Bid No."..."

Dear Sir,

I have read the clause regarding restrictions on procurement from a bidder of a country which shares a land border with India. I certify that this bidder is not from such a country of if from such a count has been registered with competent authority is hereby certify that this bidder fulfils all requirement in this regard and is eligible to be considered. (Where applicable, evidence of valid registration by the competent authority shall be attached).

Date:

Signature.....

(Authorised signatory)

Place:

Full Name.....

Designation.....

Seal:

Integrity Pact
(To be submitted on Bidder's Letterhead)

To
The Tender Inviting Authority,
Employer Address

Subject: Submission of Integrity Pact for Bid No."..."

It is confirmed and declared that we, or any of our associate, have not been engaged in any fraudulent and corrupt practice and that no agent, middleman or any intermediary has been, or will be, engaged to provide any services, or any other items of work related to the award and performance of this contract and no agency commission or any payment which may be construed as an agency commission has been, or will be, paid and that the Bid price will not cover any such amount.

Date:	Signature.....
	(Authorised signatory)
Place:	Full Name.....
	Designation.....
	Seal:

END OF DOCUMENT

TERMS OF REFERENCE

- ❖ This TOR shall replace Sl. No. 4 (Scope of Work) of the Service Level Agreement (SLA)
- ❖ The Consultant shall refer to this TOR to understand and deliver the scope of work for this assignment.

INDEPENDENT SAFETY AUDIT

Name of the Work: Consultancy Assignment for Conducting an Independent Safety Audit to Enhance Compliance with Safe Systems of Work in Chennai Metro Rail Limited (CMRL) Phase-2 Civil and Track Projects – Section A

1. BACKGROUND

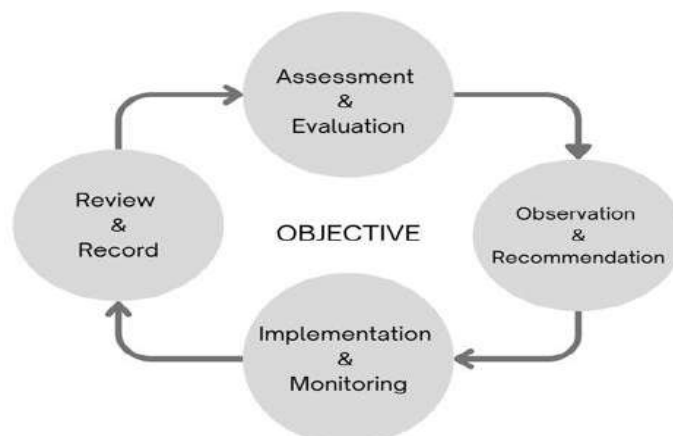
Chennai Metro Rail Limited (CMRL) is a joint venture of the Government of India (GoI) and the Government of Tamil Nadu (Go TN) for construction of Metro Rail Project Phase I & Phase 2 in Chennai city, with financial support from the funding agencies like JICA/ADB/NDB/AIIB.

Phase 1 spans a network of 54.1 kilometers, comprising Corridor 1 (Wimco Nagar to Airport, 32.1 km) and Corridor 2 (Chennai Central to St. Thomas Mount, 22.0 km). Presently, 54.1 km of Phase 1 is operational.

Phase 2 consists of three corridors covering a combined length of 116.1 km: Corridor 3 (Madhavaram to Siruseri, 45.4 km), Corridor 4 (Lighthouse to Poonamallee, 26.1 km), and Corridor 5 (Madhavaram to Sholinganallur, 44.6 km). The construction of all the corridors of phase 2 is in full swing.

2. OBJECTIVE:

The objective of this independent assessment is to systematically review, assess, and evaluate the implementation of approved work Method Statements through structured site observations. The assessment aims to identify gaps, deviations, and non-compliances, record audit evidence and associated consequences, and provide objective, practical recommendations for corrective and preventive action. The implementation of these recommendations shall be monitored and verified to ensure continual improvement in compliance, safety, and performance, in alignment with the PDCA (Plan–Do–Check–Act) cycle.



3. SCOPE:

The scope of the audit covers CMRL Phase–II Civil and Track Projects, comprising two (2) elevated packages, eight (8) tunnel and underground station packages, one (1) depot civil package, and two (2) track packages, namely C5-EV-03, C3-EV-01, C3-TU-01, C3-TU-02, C3-UG-01, C3-UG-02, C3-UG-03, C3-UG-04, C3-UG-05, C5-UG-06, C5-DPT-02, C3-ATW-09, and C5-ATW-12, as detailed in Annexure-A of the Additional Terms and Conditions of contract (ATC).

- **Auditing adherence to and implementation of** approved Work Method Statements, and **identifying deficiencies and deviations** from them, in the context of applicable IS Codes related to quality, temporary structural designs, CMRL's Occupational Health & Safety (OH&S) Conditions of Contract (refer ATC for CMRL OH&S COC), and relevant national statutes and rules, including the Building and Other Construction Workers (BOCW) Act, 1996, and Tamil Nadu BOCW Rules, 2006, etc.
- **Reviewing, simplifying, and recommending improvements** to Work Method Statements, Standard Operating Procedures, and safety protocols, wherever required, to enable effective, practical, and safe implementation at site, in line with CMRL safety requirements.

4. DUTIES AND RESPONSIBILITIES OF THE CONSULTANT

Below are the duties and responsibilities of the consultant but not limited to.

4.1 The Consultant shall form a multi-disciplinary team for this assignment. The independent Consultant team shall be staffed with an adequate number of personnel with relevant experience in a similar scope of work. The detailed requirements for Key and Non-Key Personnel are provided in Section 10 of this TOR.

4.2 The Consultant shall maintain an office in Chennai, and all Key and Non-Key Personnel shall be available in Chennai to provide continuous input throughout the full duration of the assignment.

4.3 The Consultant shall ensure that the same Key and Non-Key Personnel are not deployed concurrently for any other contract, either within CMRL or with any external organization, during the entire duration of this contract

4.4 The Consultant shall meet with CMRL/GC/Contractor's authorized officials of contract packages to understand the present work activities and its approved work method statements. The Consultant shall collect relevant documents, data, and insights from CMRL, GC, Contractor, and other stakeholders; analyse current processes, operations, and systems to identify issues or opportunities for improvement; and prepare a detailed roadmap with timelines, milestones, and deliverables.

4.5 The Consultant shall be available at the site for audits well before the commencement of work, based on the requirements of day/night shifts. The Consultant shall also share a detailed work schedule with CMRL/GC, including weekly and monthly plans for the contract period.

4.6 The Consultant shall share a detailed monthly work schedule in consultation with CMRL/GC. The selection of Method Statements and contract packages for review and the planning of site audits shall be carried out in consultation with CMRL and GC, based on the nature and progress of works.

4.7 The Consultant shall assess the contractor's understanding and awareness of the approved safety procedures, including Work Method Statements, using a checklist or questionnaire discussed with CMRL/GC prior to the commencement of work at site.

4.8 The team of Key and Non-Key Personnel shall, as professional and experienced consultants, accept full responsibility for the accuracy, correctness, and technical merit of the services performed.

4.9 All the additional incidental services related to the preparation of the report highlighting issues during the service should be deemed to be included in the scope of services of consultants. No extra payment shall be made for such incidental services. Additional incidental services for the consultant may include but are not limited to the below:

- a) Site inspection carried out on day/night shift-based works requirements,
- b) Analysing specific construction processes and operations to uncover unique risks,
- c) Compliance checks and regulatory review,
- d) Documenting and developing detailed action plans to address safety violations or non-conformances, if any,
- e) Investigating any past incidents or accidents on the construction site and suggesting safety improvements, if any,
- f) Identifying gaps in current work method statements, safety protocols, safety procedures, and training programs, and recommending tailored suitable suggestions to enhance safety standards in CMRL projects based on findings,
- g) Providing post-report follow-up to monitor the implementation of safety recommendations,
- h) Scheduling additional safety inspections and monitoring to verify that corrective actions have been taken or implemented,
- i) Timely coordination with concerned CMRL / General Consultant (GC) / Contractor's representatives during the contract period.

4.10 The Consultant shall proactively maintain regular and effective communication with CMRL/GC and the Contractors, take responsibility for identifying and addressing issues in a timely manner, and keep CMRL informed of progress and resolutions throughout the contract period.

4.11 The Consultant shall remain updated with applicable industry standards, IS Codes, statutory regulations, and relevant innovations, and shall incorporate such knowledge,

wherever appropriate, to provide relevant and practical advice and recommendations to CMRL.

5. COORDINATION MEETINGS

The following coordination meetings shall be conducted to ensure effective implementation and monitoring of the consulting services:

5.1. Regular fortnightly coordination meetings shall be held during the contract period, wherein **CMRL shall review the progress** of work and other relevant aspects of the assignment. These meetings shall be conducted at CMRL Headquarters and/or CMRL project site offices, as decided by CMRL.

5.2 In addition to the fortnightly coordination meetings, the Consultant shall conduct **monthly package-level coordination meetings** with the concerned **GC and Contractor's representatives** to review site-specific safety observations, compliance status, and the implementation of audit recommendations, with the **concerned CMRL officials duly informed**.

5.3. The Consultant **shall prepare and submit comprehensive progress reports** summarizing audit findings, recommendations, and implementation status, and shall make **formal presentations to CMRL** during coordination meetings highlighting key observations, issues, and proposed actions. The Consultant shall ensure continuous communication to maintain alignment with CMRL objectives.

5.4. The Consultant **shall record the Minutes of Meeting (MoM)** for all coordination meetings and submit the same to CMRL for review and concurrence within two (2) to three (3) working days from the date of the meeting. Approved MoMs' shall be maintained by the Consultant as official records for future reference.

5.5. All Key Personnel shall be physically present during coordination meetings. Non-Key Personnel shall attend such meetings as and when required by CMRL, based on the agenda and subject matter.

6. ACTIVITIES & TASKS TO BE PERFORMED BY THE CONSULTANT AND THEIR TIMING:

The activities and tasks to be performed by the Consultant shall include, but not be limited to, the following:

6.1 The Consultant shall collaborate with CMRL/GC to understand the packages, scope of works, and necessary information for the preparation of audit weekly/monthly plans.

6.2 The Consultant shall review and familiarize themselves with the approved Work Method Statements prior to site assessment, analyse the processes involved, and evaluate potential risks and challenges associated with the work activities.

6.3 The Consultant shall coordinate with the respective Contractor representatives to collect the required documents and reports and shall keep the concerned CMRL/GC officials duly informed.

6.4 The Consultant shall coordinate with the respective CMRL/GC Safety or Civil Managers for the selection of activities or locations to be audited and the allocation of man-days. The same shall be formally communicated to CMRL through official email.

6.5 The Consultant shall notify the concerned Contractor at least two (2) working days prior to the scheduled audit and obtain details of ongoing activities from the Contractor's team.

6.6 The Consultant shall seek the physical presence of the Contractor's personnel by notifying the concerned package representatives of CMRL/GC/Contractor in advance and shall conduct the audit accordingly.

6.7 The Consultant shall verify the sequence of work, control measures, and work procedures implemented at site against the approved Method Statements. Where alternative methodologies are adopted by the Contractor, the Consultant shall verify their adequacy and approval status with CMRL/GC.

6.8 The Consultant shall evaluate whether the control measures and procedures specified in the Method Statements are practicable, effective, and designed by competent persons to prevent accidents. Any deficiencies in implementation shall be formally reported, including reasons and areas of deficiency.

6.9 The Consultant shall assess the adequacy of Method Statements in addressing practical difficulties and challenges encountered by workers during execution.

6.10 The gaps observed in the implementation of Method Statements shall be assessed against the requirements specified in the approved Method Statements and CMRL's Conditions of Contract. **Any deficiencies or inadequacies in the Method Statements shall be corroborated with reference to applicable statutory provisions, relevant IS/BS Codes, and recognized national and international safety standards.**

6.11 The Consultant shall verify that all temporary structures at site are erected and installed strictly in accordance with approved drawings and specifications, ensure that approved design drawings, design calculations (including load calculations), and stability checks are available for all temporary structures, and verify that such temporary works have been designed, reviewed, and approved by a competent authority and certified, wherever applicable, prior to and during execution.

6.12 The Consultant shall verify the quality and type of materials used at site in accordance with the specifications provided in the approved Method Statements and applicable IS Codes.

6.13 The Consultant shall review the roles and responsibilities of personnel as defined in the organization chart for each activity covered under the Method Statements.

6.14 The Consultant shall prioritize the audit of critical activities, particularly those where public safety and life safety are at risk, in concurrence with the concerned CMRL and GC representatives.

6.15 The Consultant shall ensure that any potentially high-risk or imminent danger observations, particularly those affecting public safety or life safety, are immediately communicated to the concerned Contractor and GC for prompt corrective action, with simultaneous intimation to CMRL.

6.16 The Consultant shall review revised Method Statements and risk assessments prepared following any fatal accident, major injury, or dangerous occurrence to verify the adequacy of corrective and preventive measures implemented.

6.17 The Consultant shall review working hours, overtime records, health records, training records, and other relevant details of workers employed by Contractors in line with CMRL requirements and applicable statutory provisions.

6.18 This assignment requires full-time availability of Key and Non-Key Personnel. The Consultant shall maintain attendance records of deployed personnel, which shall be verified by the appropriate authorities.

6.19 The Consultant shall inform the respective package CMRL/GC Safety Managers regarding the auditors' attendance and presence at site by submitting required documents and time-stamped photographs through official WhatsApp groups and/or official email, as specified by CMRL.

6.20 The Consultant shall ensure that **all observations recorded with agreed timeline for rectifications** during site visits are **documented in a daily auditor's observation sheet** (in the format agreed with CMRL/GC), are duly discussed, communicated, and signed off by the concerned Contractor, and verified by representatives of CMRL/GC.

6.21 The Consultant shall provide all deliverables as specified in Section 8 of this TOR within the prescribed timelines.

6.22 The Consultant shall maintain proper records of all report submissions and communications, including email confirmations, acknowledgements, and submission receipts, and shall maintain a consolidated register of such records for verification by CMRL.

7. Man-Day Allocation, Attendance Certification and Payment Linkage

7.1. For the purpose of monitoring deployment of Key and Non-Key Personnel and for certification of payments, the Consultant shall maintain and submit **monthly man-day attendance records** indicating the allocation of professional time spent on various audit activities.

7.2. The indicative distribution of man-days for each Key and Non-Key Personnel shall be as follows:

- **15% of total man-days** – Review of Method Statements, Risk Assessments, Safety Procedures and related documentation.

- **70% of total man-days** – Site visits, field audits, physical verification, interaction with Contractor / GC / CMRL representatives (including day/night shifts as applicable).
- **15% of total man-days** – Preparation of reports (GIR, IM, Monthly Interim reports), follow-up actions, and participation in coordination / review meetings with CMRL, GC and Contractors.

7.3. Each Key and Non-Key Personnel shall deploy a **minimum of 22 man-days per month**, which shall be mandatory for payment consideration.

7.4. The Consultant shall submit **monthly attendance sheets** in the agreed format, duly certified by the Consultant and countersigned by the authorized representatives of CMRL / GC, clearly indicating:

- Name and designation of Key and Non-Key personnel
- Number of man-days deployed
- Activity-wise allocation of man-days
- Dates and locations of site visits / meetings

7.5. Payments shall be linked to the submission and approval of the monthly man-day attendance records along with the corresponding deliverables, and non-compliance with the minimum man-day requirement or improper allocation may result in proportionate deduction or withholding of payment as per contract provisions.

8. DELIVERABLES:

Sl. No	Deliverables	Cumulative Timeline (from the date of commencement of assignment)
1.	Gap Identification & Recommendation (GIR) Report	Every 7 days
2.	Implementation and Monitoring Report – IM Report	Every 14 days
3.	Monthly Interim Report (GIR + IM Report)- MI Report	Every 30 days
4.	Draft Final report	350 days
5.	Final Report	365 days

* The commencement date of assignment shall be **14 days** from the date of issuance of LOA

8.1 Gap Identification & Recommendation (GIR) Report:

The Gap Identification & Recommendation (GIR) Report shall include, but not be limited to, the following:

8.1.1. The report shall **contain detailed observations on deviations** in the implementation of approved Work Method Statements (MS) observed during site visits, clearly indicating the specific activity, location, and package reference.

8.1.2. The Consultant shall **identify and document inadequacies or deficiencies** within the Method Statements themselves while evaluating their implementation, including missing controls, impractical procedures, or insufficient risk mitigation measures.

8.1.3. The Consultant shall ensure that all observations and recommendations included in the **GIR Report are in accordance with applicable National and International standards and codes of practice** related to Safety, quality, temporary structural designs, underground and tunnelling construction, etc., applicable statutory requirements (including but not limited to the Building and Other Construction Workers (BOCW) Act, 1996 and Tamil Nadu BOCW Rules, 2006), and CMRL's Occupational Health & Safety (OH&S) Conditions of Contract.

8.1.4. The **recommendation** provided for each identified gap shall be objective, clear, **practical, and implementable at site**.

8.1.5. The Consultant shall ensure that all recommendations **are made with due reference to the relevant IS/BS Codes of Practice, applicable national statutory rules and regulations**, and CMRL's Occupational Health & Safety (OH&S) requirements.

8.1.6. All observations recorded during site visits while evaluating the implementation of Method Statements shall be systematically documented and supported with verifiable audit evidence, including photographs, relevant documents, records, and interactions with the Contractor's site personnel, ensuring traceability of each observation.

8.2 Implementation and Monitoring (IM) Report:

The Implementation and Monitoring (IM) Report shall include, but not be limited to, the following:

8.2.1. The report shall clearly indicate the status of implementation and adherence to each recommendation issued in the GIR Report.

8.2.2. The Consultant shall verify the implementation status at site and update the status in the IM Report (for the purpose of getting implementation status, the consultant shall coordinate with the respective GC/Contractor's safety representatives), supported with verifiable photographic evidence, relevant documents, and records, clearly demonstrating the closure or progress of each recommendation.

8.2.3. Each recommendation in the IM Report shall be assigned a clear implementation status (Closed / Partially Closed / Open), along with the date of verification."

8.2.4. The report shall provide a quantitative summary of risks mitigated and the corresponding safety benefits (value addition) realized by CMRL as a result of implementing the audit recommendations. Risk mitigation shall be expressed in numerical terms, based on risk assessments, including a comparison of risk levels before and after implementation of recommendations by the audit team

8.3 Monthly Interim Report (MI Report):

The Monthly Interim Report shall include, but not be limited to, the following:

8.3.1. The report shall provide comprehensive documentation of the **GIR and IM Reports**, including detailed explanations of how each recommendation contributes to improvement in project safety, quality and temporary structures design & constructions.

8.3.2. The report shall include details of the Method Statements reviewed and verified at project sites during the reporting period. **A minimum of twelve (12) Method Statements per month**, related to ongoing high-risk and critical activities of the awarded **Section A packages including system contract packages also if required**, shall be reviewed, and the implementation status shall be physically verified at site during the audit in concurrence with CMRL.

8.3.4. CMRL may issue special assignments, apart from the assigned scope of work, **based on project requirements**. Such assignments shall be considered part of the regular assignment at no extra cost. The Vendor shall accommodate these assignments along with the regular ongoing assignments and shall submit the required reports to CMRL.

8.3.5. The report shall provide a **summary of site visits and audit activities** carried out by the Key and Non-Key Personnel during the reporting period.

8.3.6. Summary of Compliant Observations from the Site Audit: The report shall provide a detailed summary of all compliant observations recorded during the site audit. Each compliant observation shall be documented sub-activity-wise, in accordance with the approved Method Statement, and shall be supported with photographic evidence including date, time, and location details, along with relevant reference documents demonstrating adherence to specified procedures and standards.

8.3.7. Summary of Deviations and Deficiencies: The report shall summarize all **potentially high-risk deviations and deficiencies** identified during the site audit. These findings shall be recorded in line with the approved Method Statement and categorized sub-activity-wise for clarity. **High-risk deviations shall be clearly highlighted for priority corrective action**. Each potentially high-risk deviation or deficiency shall be supported with photographic evidence (with date, time, and location details) and relevant documentary records to facilitate accurate assessment and follow-up actions.

8.3.8. Summary of Critical Observations Not Covered in the Method Statement: The report shall highlight any **critical sequences, activities, or observations** identified during the site audit that are **not addressed or covered** in the approved

Method Statement. Such observations shall be documented to identify gaps in the existing Method Statements and shall be supported with **photographic evidence and relevant documentation**.

8.3.9. All potentially high-risk observations included in the Monthly Interim (MI) Report shall clearly reference the corresponding GIR and IM Report numbers to ensure traceability and continuity of audit findings and corrective actions.

8.4 Final Report:

The Final Report shall present a consolidated summary of the Independent Safety Audit conducted to enhance safety compliance in CMRL Phase-II Civil and Track Projects (Section-B), covering the entire contract period, and shall include, but not be limited to, the following:

8.4.1. Summary of Audit Coverage:

A consolidated summary of audit activities conducted during the assignment, including the contract packages covered, types of activities audited, duration of audits, and deployment of Key and Non-Key Personnel.

8.4.2. Consolidation of Monthly Interim Reports:

A compilation and summary of all approved Monthly Interim Reports (IRs) submitted during the contract period, ensuring continuity and traceability of observations, recommendations, and follow-up actions.

8.4.3. Overall Gap Analysis and Trends:

An overall analysis of gaps observed during the audit, highlighting recurring deviations, improvement trends in Method Statement compliance, and areas requiring continued management attention. Trend analysis shall be supported with qualitative and quantitative data.

8.4.4. Key Observations and Recommendations:

A summary of significant observations, critical safety issues, and key recommendations, categorized activity-wise and package-wise, with reference to applicable Method Statements, OHS&E Conditions of Contract, and statutory requirements.

8.4.5 Implementation Status and Effectiveness:

A summary of the implementation status of audit recommendations, including the percentage implemented, pending actions with reasons, and the effectiveness of implemented controls, supported with relevant photographic and documentary evidence.

8.4.6 Risk Mitigation and Safety Performance:

An assessment of overall risk reduction and improvement in safety performance achieved through the audit, including reduction in risk levels and enhancement of safety

controls and work practices, quantified using the approved risk assessment methodology.

8.4.7 KPI Performance Summary:

A consolidated summary of performance against the defined KPIs, including Method Statement Compliance Rate, Audit Observation Closure Rate, Risk Mitigation Effectiveness, and timeliness of implementation of recommendations.

8.4.8 Value Addition and Way Forward:

A summary of value addition achieved for CMRL through the audit, key lessons learned, and practical recommendations for continuous safety improvement and ongoing monitoring in CMRL Phase-II projects.

8.4.11 Mandatory Annexures to the Final Report:

The Final Report shall include the following annexures. Non-submission or incomplete submission of any annexure may result in withholding of the final payment until compliance:

Annexure A: Audited Method Statements

- Complete list of Method Statements audited during the contract period
- Classification by package and activity
- Dates of review and site verification

Annexure B: Consolidated GIR Register

- Summary of gaps, deviations, and deficiencies
- Reference to GIR Report numbers and dates
- Status of each observation (Closed / Partially Closed / Open)

Annexure C: Implementation & Monitoring (IM) Register

- Implementation status of recommendations
- Evidence references (photograph/document numbers)
- Dates of closure and verification

Annexure D: Risk Assessment & Mitigation Summary

- Risk scores before and after implementation
- Activity-wise risk reduction analysis
- Basis of risk scoring adopted by the audit team

Annexure E: KPI Performance Summary

- KPI-wise performance results and trend analysis

- Final consolidated KPI outcomes

Annexure F: High-Risk & Critical Observation Register

- List of high-risk and critical observations
- Corrective and preventive actions taken
- Final risk status

Annexure G: Attendance & Man-Day Records

- Monthly attendance records of Key and Non-Key Personnel
- Activity-wise man-day allocation
- Certification and countersignature by CMRL / GC

Annexure H: Minutes of Meetings (MoM)

- Approved Minutes of coordination and review meetings
- Action points and compliance status

8.5 Report Submission Procedures:

8.5.1 The Consultant shall finalize the reporting formats for the **GIR, IM & MI Reports** in coordination with CMRL **prior to the commencement of any official report submission**. All reports specified in Section 8 shall be submitted **within the prescribed timelines**, in soft copy and/or hard copy, **as required by CMRL**, and in the approved formats (such as PDF, Word, Excel, etc.).

8.5.2 The Consultant shall submit the **GIR and IM Reports** to the concerned **GC representatives**, with a copy to CMRL, to obtain compliance status and details of pending actions from the concerned Contractors. The **GIR and IM Reports shall be submitted within one (1) to two (2) working days** after the timelines specified in the Deliverables Clause (Section 8).

8.5.3 The Consultant shall coordinate with the concerned **GC representatives** to obtain compliance confirmations on the GIR observations and shall submit the **updated status reports** to CMRL.

8.5.4 The Consultant shall submit the **Monthly Interim Report (MI Report)** and the **Final Report** to CMRL, along with all supporting documents, **within three (3) working days** after the timelines specified in the Deliverables Clause (Section 8).

9. KEY PERFORMANCE INDICATORS (KPIs):

The performance of the Consultant shall be monitored and evaluated using the following Key Performance Indicators (KPIs):

9.1. Audit Observation Closure Rate:**Definition:**

Measures the effectiveness of implementation and closure of audit recommendations issued by the Consultant.

Formula:

$$\text{Observation Closure Rate (\%)} = \left(\frac{\text{Number of audit observations closed}}{\text{Total number of audit observations issued}} \right) \times 100$$

Measurement Source:

- Implementation and Monitoring (IM) Reports

9.2. Risk Mitigation Effectiveness Index:**Definition:**

Assesses the effectiveness of audit recommendations in reducing safety risks associated with construction activities.

Formula:

$$\text{Risk Mitigation Effectiveness (\%)} = \left(\frac{\text{Total risk score after implementation}}{\text{Total risk score before implementation}} \right) \times 100$$

(Lower percentage indicates better risk mitigation.)

Measurement Source:

- Risk assessments carried out by the audit team
- Implementation and Monitoring (IM) Reports

9.3. Timely Implementation Rate of Recommendations:**Definition:**

Measures the proportion of audit recommendations implemented within the agreed timelines.

Formula:

$$\text{Timely Implementation Rate (\%)} = \left(\frac{\text{Number of recommendations implemented within agreed timeline}}{\text{Total number of recommendations issued}} \right) \times 100$$

Measurement Source:

- Implementation and Monitoring (IM) Reports
- Follow-up audit records

9.4. High-Risk Observation Reduction Rate

Definition:

Measures the reduction in high-risk or critical safety observations over successive reporting periods.

Formula:

$$\text{High-Risk Observation Reduction (\%)} = \left(\frac{\text{Number of high-risk observations in current period}}{\text{Number of high-risk observations in previous period}} \right) \times 100$$

(A decreasing percentage over time indicates improvement.)

Measurement Source:

- Interim Reports
- GIR trend analysis

9. NUMBER OF COPIES FOR FINAL REPORT SUBMISSION

Sl. No.	Description	No. of copies required
A	Final report signed sealed by an authorized signatory – Hard Bonded Photo Print A4/A3 to a readable size	05 no's each
B	All other documents & reports – Bonded A4/A3 to a readable size	05 no's each

10. QUALIFICATION AND EXPERIENCE OF PERSONNEL (KEY & NON-KEY PERSONNEL):

10.1. The qualification and experience requirements for the Professional key and non-key personnel as given below are the CMRL's requirement. The applicant is advised to prepare the CVs of their proposed Personnel (Key & Non-Key)

specifically highlighting their qualifications and experience in the relevant areas of expertise. The CVs should also highlight the proposed Professional Staff's academic qualifications, professional qualifications & certifications, etc. as well as their experience in conducting similar audits for large infrastructure construction projects preferably metros (Elevated / Underground) / railways / bridges & road construction projects. The CVs of Personnel (Key & Non-Key) to be deployed for the project shall be submitted to CMRL for approval before commencement of assignment. No Personnel engaged in the assignment shall be replaced without prior intimation to CMRL. Bidders shall fill in Form Tech-3 (Curriculum Vitae of proposed Key Personnel) as part of the Evaluation and Qualification Criteria for technical bid evaluation.

KEY PERSONNEL:

Lead Auditor – 1No		
I	Educational qualification	
	Essential	BE/BTech/ME/MTech in civil or mining or structural or design.
II	Professional qualification	
		Certified Lead Auditor ISO 9001 or 45001, etc.
III	Essential experience	
	a) Total professional experience	Min.15 years for graduate.
	b) Experience in relevant works	Minimum a total 08 years of experience in construction, conducting audits, providing consulting services for large infrastructure construction projects preferably in metros (Elevated & Underground) / railways / bridges & road construction projects.
IV	Age limit	
		Not exceeding 65 years on the date of submission of Proposal

Deputy - Lead Auditor – 1 No		
I	Educational qualification	
	Essential	BE/BTech/ME/MTech in civil or mechanical or mining or structural or design. Preferably Civil department M.E / MTech in industrial safety engineering / RLI - PGDIS/ PG diploma in industrial safety approved by state technical board
II	Professional qualification	
	Essential	Certified Lead Auditor ISO 9001 or 45001 or CSP or any other equivalent safety certification on CMRL approval

III	Essential experience	
	a) Total professional experience	Min. 12 years for graduate.
	b) Experience in relevant works	Minimum a total 06 years of experience in conducting safety audits, providing safety consulting services for large infrastructure construction projects preferably in metros (Elevated & Underground) / railways / bridges & road construction projects.
IV	Age limit	Not exceeding 65 years on the date of submission of Proposal

NON-KEY PERSONNEL:

Auditor – 3 Nos		
I	Educational qualification	
	Essential	BE/BTech/ME/MTech/Diploma in civil or mechanical or mining or structural or design. Preferably Civil & Mechanical department (Two auditor in civil & One auditor in mechanical educational background)
II	Professional qualification	
	Added advantage	Lead auditor ISO 45001 or ISO 9001 or any safety / quality certification etc.
III	Essential experience	
	a) Total professional experience	Min. 07 years for graduate / min. 9 years for diploma
	b) Experience in relevant works	Minimum a total 04 years of experience in construction, conducting audits, providing consulting services for large infrastructure construction projects preferably in metros (Elevated & Underground) / railways / bridges & road construction projects.
IV	Age limit	Not exceeding 36 years on the date of submission of Proposal

Note:

- 10.1.1. Where a potential candidate has previously worked for large infrastructure construction projects preferably in metros / railways / bridges & road construction projects and does not possess the qualifications and or the necessary experience as listed in above table **for the Non-Key Personnel role**, CMRL may accord his approval upon a successful interview of the candidate on an ad hoc basis including age relaxation.
- 10.1.2. The Key/ Non key personnel must have practical understanding of **Building and Other Construction workers (BOCW) Act** and Rules, other applicable statutory requirements, health / medical and welfare of workmen, construction



Consultancy Assignment for Conducting an Independent Safety Audit to Enhance Compliance with Safe Systems of Work in Chennai Metro Rail Limited (CMRL) Phase-2 Civil and Track Projects – Section A

hazards and its prevention & control, traffic management, electrical safety, rigging, and safety of construction equipment.

***** END OF DOCUMENT *****

Sanjay Ram
778
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BUREAU VERITAS (INDIA) PRIVATE LIMITED
MUMBAI

PAYMENT TERMS & CONDITIONS

1. Payment Terms:

- 1.1. 100% payment shall be made in accordance with the payment schedule defined below and shall be released **within twenty-eight (28) days** from the date of issuance of the **Consignee Receipt-cum-Acceptance Certificate (CRAC)** and successful online submission of invoices through GeM, subject to approval by CMRL.
- 1.2. Payments shall be milestone-based and linked to **deployment of Key and Non-Key personnel (minimum 22 man-days per month)** and **submission & approval of reports**, as detailed below:

2. Payment Schedule:

2.1. Monthly Payments (During Contract Period for 11 months):

SI No	Description	Percentage of the amount to be paid
2.1. A	Submission and approval of monthly attendance for Key & Non-Key Personnel with minimum 22 man-days per personnel (as per Section 7 of TOR on Man-Day Allocation & Attendance Certification)	5 % of accepted contract value per month
2.1. B	Submission and approval of Monthly Interim Report (GIR + IM Report)	3 % of accepted contract value per month
	Total Monthly Payment	8 % of accepted contract value per month

2.1.2. Monthly payment shall be released only upon approval of both Part 2.1.A and Part 2.1.B. Failure to meet the minimum man-day requirement shall result in proportionate withholding of payment. The withholding amount shall be released upon fulfilment of the above requirements.

2.2. Final Payment (for 12th month):

SI No	Description	Percentage of the amount to be paid
2.2. A	Submission and approval of monthly attendance for Key & Non-Key Personnels with minimum 22 man-days per personnel (as per Section 7 of TOR on Man-Day Allocation & Attendance Certification)	6.5 % of accepted contract value per month
2.2. B	Submission and approval of Final Report	5.5 % of accepted contract value per month

	Total Final Payment	12 % of accepted contract value per month
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3. General Conditions Applicable to All Payments:

3.1 Monthly attendance sheets shall be submitted in the prescribed format, duly certified by the Consultant and countersigned by CMRL / GC.

3.2 Payments shall be subject to:

3.2.1. Compliance with Man-Day Allocation, Attendance Certification and Payment Linkage clause

3.2.2. Acceptance / approval of reports by CMRL

3.2.3. Submission of complete supporting documents

3.3 Taxes shall be paid as applicable and as per GeM guidelines.

Tender Inviting Authority: CHENNAI METRO RAIL LIMITED							
Name of Work: Consultancy Assignment for Conducting an Independent Safety Audit to Enhance Compliance with Safe Systems of Work in Chennai Metro Rail Limited (CMRL) Phase-2 Civil and Track Projects – Section A							
Name of the Bidder/ Bidding Firm / Company :							
PRICE SCHEDULE: (DOMESTIC TENDERS - RATES ARE TO GIVEN IN RUPEES (INR) ONLY) (This BOQ template must not be modified/replaced by the bidder and the same should be uploaded after filling the relevant columns, else the bidder is liable to be rejected for this tender. Bidders are allowed to enter the Bidder Name and Values only) 1. If there is any increase or decrease in the GST due to change in applicable law during the currency of contract, such additional or reduced cost shall be paid by or credited to the Client. Note: 1.Price to be filled by bidder without GST 2.The Evaluation will be done based on the cost quoted excluding GST. 3. GST Amount will be reimbursed separately on submission of Proof of discharge of liability.							
S.No	Description	Quantity	Units	Total Amount Without GST In Figures To be entered by the Bidder in Rs.	GST in %	Total Amount Including GST Rs.	TOTAL AMOUNT In Words
1	Consultancy Assignment for Conducting an Independent Safety Audit to Enhance Compliance with Safe Systems of Work in Chennai Metro Rail Limited (CMRL) Phase-2 Civil and Track Projects – Section A (Fee to be quoted as Lump-sum amount excluding GST)	1	JOB		18%	-	

Note :The quoted price in above price schedule shall be inclusive of the following all assignment related costs but not limited to:

a) The consultants will have to make their own arrangements for the residential / office accommodation or any other facilities at site, transport / TA / DA and accommodation & related expenses of their staff deployed for the present assignment including visiting various offices, and other places for meetings, data collection, presentations, public consultation during field visit, public hearing and discussions, the remuneration of the experts, secretarial staff, their salary, allowances & overhead expenditure, all required insurances etc.

b) Bid price quoted would deem to have included all the incidental costs including cost of data collection from various line departments, field survey, all documents and reports etc. that would be required to-be prepared and submitted by the consultants during entire course of the present assignment.

c)All duties, taxes, royalties and other levies payable by the Consultant under the Contract, or for any other cause, shall be included in the total Bid price except for Goods and Service Tax (GST).

Cost breakup on Quoted amount excluding GST including all the requirements mentioned in TOR and other tender requirements		
S.NO	Description	Amount in Rs (Excluding GST)
1	Remuneration for Key person 1 (Lead Auditor)	
2	Remuneration for Key person 2 (Deputy - Lead Auditor)	
3	Remuneration for Non - Key person 1 (Auditor 1)	
4	Remuneration for Non - Key person 2 (Auditor 2)	
5	Remuneration for Non - Key person 3 (Auditor 3)	
6	Others (Consultant's obligations like conveyance, overhead cost, documentation,etc.)	
	Total	

Section – 12

Form TECH-1: Technical Proposal Submission Form



Form TECH-1: Technical Proposal Submission Form

Date: 27.01.2026

GeM Bid No.: **GEM/2026/B/7074523** Date: **05-01-2026**

Project/Assignment : **Consultancy Assignment for Conducting an Independent Safety Audit to Enhance the Compliance with Safety Systems of Work in Chennai Metro Rail Limited (CMRL) Phase-2 Civil & Track Projects – Section A**

To,
Chief General Manager (P&D)
Chennai Metro Rail Limited,
METROS, Anna Salai, Nandanam, Chennai 600035.

Dear Sir/ Madam:

We, the undersigned, offer to provide the consulting Assignment for **Consultancy Assignment for Conducting an Independent Safety Audit to Enhance the Compliance with Safety Systems of Work in Chennai Metro Rail Limited (CMRL) Phase-2 Civil & Track Projects – Section A** in accordance with your Request for Proposals **GEM/2026/B/7074523** including addenda issued **GEM/2026/B/7074523** Date: **05-01-2026** and our Proposal.

We are hereby submitting our Proposal which includes [insert “this Technical Proposal, and a Financial Proposal sealed in a separate envelope”].

We hereby declare that:

- (a) All the information provided and statements made in this Proposal are true and we accept that any misinterpretation or misrepresentation contained in this Proposal may lead to our disqualification by the Employer.
- (b) Our Proposal shall be valid for 90 days from the date of Bid submission and remain binding upon us for the period of time specified.
- (c) We have no conflict of interest in accordance with clause as per NIT.
- (d) We understand that the Employer is not bound to accept any Proposal that the Employer receives.
- (e) We hereby certify that we have taken steps to ensure that no person acting for us or on our behalf will engage in any type of fraud and corruption.
- (f) We hereby certify and declare that if awarded with this assignment “no form of child labor or bonded labor shall be employed anywhere in the supply chains”.



**BUREAU
VERITAS**

- (g) We hereby certify that the proposal submitted by me / us is strictly in accordance with the terms, conditions, Terms of Reference etc. as contained in the RFP document, and it is further certified that it does not contain any deviation to the aforesaid documents.

We remain,

Yours sincerely,

Thanking you,

Yours very truly,

For Bureau Veritas (India) Pvt. Ltd.

.....
Surajit Sankar Pan

Director - B&I, CIF SAR

Date: 27.01.2026

Regd. Office: 72 Business Park, Marol Industrial Area,
Opposite Seepz Gate No. 2, MIDC Cross Road "C",
Andheri - (East), Mumbai - 400 093. INDIA
Tel: +91 22 62742000 • Fax: +91 22 62742008
E-mail: bni.mumbai@bureauveritas.com

Section – 13

Form FIN-1: Financial Proposal Submission Form



Form FIN-1: Financial Proposal Submission Form

Date: 27.01.2026

GeM Bid No.: **GEM/2026/B/7074523** Date: **05-01-2026**

To,
Chief General Manager (P&D)
Chennai Metro Rail Limited,
METROS, Anna Salai, Nandanam, Chennai 600035.

Dear Sir/ Madam:

We, the undersigned, offer to provide the consulting Assignment for **Consultancy Assignment for Conducting an Independent Safety Audit to Enhance the Compliance with Safety Systems of Work in Chennai Metro Rail Limited (CMRL) Phase-2 Civil & Track Projects – Section A** for Chennai Metro Rail Limited in accordance with your Request for Proposals dated 27.01.2026 and its addendum.

We are hereby submitting our Financial Proposal for the sum of **[QUOTED ONLINE].**

Only]. We hereby declare that

- (a) Our Financial Proposal shall be binding upon us subject to the modifications, if any, up to expiration of the validity period of the Proposal as mentioned in
- (b) We understand that the Employer is not bound to accept any Proposal that the Employer receives.

We remain,

Yours sincerely,

Thanking you,

Yours very truly,

For Bureau Veritas (India) Pvt. Ltd.

.....
Surajit Sankar Pan
Director - B&I, CIF SAR

Date: 27.01.2026

Regd. Office: 72 Business Park, Marol Industrial Area,
Opposite Seepz Gate No. 2, MIDC Cross Road "C",
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Section – 14

Form TECH-2: Consultant's Organization and Experience



Form TECH-2: Consultant's Organization and

Experience Consultant's Organization

Sl. no	Name of Client	Scope of work	Period of contract (From/to)	Assignment awarded Value (in Rs.)	Contract Status		Proof of work	
					Continuing (Yes/No)	Date of Completion	Copy of LOA to be attached (Yes/No)	Completed work (Client certificate to be attached)
1.	DTCP Haryana	Structural Safety Audit by conducting visual survey of multistoried buildings in Gurugram	Sep 2022 to Nov 2022	11,220,738	No	30.11.2022	Yes	
2.	Navi Mumbai International Airport	Providing Quality & Environmental Health Safety Inspection services of cut & fill works at NMIA Project	Oct 2023 to Dec 2024	4,48,69,810	No	31.12.2024	Yes	
3.	Johnson Lifts & Elevators	Third party safety audit of escalators to check their safe operational conditions as per scope of work and to enhance performance of load tests	March 2023 to April 2023	1,32,000	No	April 2023	Yes	
4.	L&T-STEEL JV Mumbai	1) Periodic HSC Audit, 2) Certification for IMS (ISO140021:2015 and OHSAS 18001:2007), 3) EHS Auditor Training 4) ISO 45001 transition Audit 5) IMS (ISO 14001:2015 and ISO 45001:2018) Recertification	Jun 2017 to Dec 2022	32,97,260	No	Dec 2022	Yes	

Bureau Veritas (India) Private Limited
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 INDIA

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 www.bureauveritas.co.in
 CIN: U74210MH2001PTC134262

Branches : Andhra Pradesh, Assam
 Chhattisgarh, Gujarat, Haryana
 Karnataka, Kerala, Madhya Pradesh
 Maharashtra, Odisha, Tamil Nadu
 Telangana, Uttar Pradesh & West Bengal





5.	L&T- STEC JV Mumbai	Testing Work	June 2018 to June 2023	8,50,530	No	June 2023	Yes	
6.	Rail Land Developm ent Authority	Appointment of Third Party Audit Agency for Quality Control/ Assurance & SHE Compliance Audit for Ongoing Station Redevelopment Projects of Indian Railways	April 2025 to Jan 2026	75,17,282	Yes	Jan 2026	Yes	Yes
7.	BIAL - Bangalore Internatio nal Airport Limited	Conduct EHS Safety Audit HSN/SAC:998349 for ongoing projects	Jan 2022 to Sept 2022	8,90,000	No	Sept 2022	Yes	
8.	Nothern Railway , Faridabad Division	Appointment of Authority Engineer for Providing Project Management Services For Major Upgradation / Redevelopment Of Faridabad Railway Station	Oct 2022 to	3,07,52,953	Yes	Partial completed April 2025	Yes	Yes Partial Physical 60% Financial 57.6%
9.	Ravi Infrabuild Projects Pvt Ltd	NH-44, NHAI's Mukarba Chowk to Panipat Section-02, From KM.29+800 to KM 54+730 in state of Haryana, under concessionaire Welspum Enterprises Limited.	Aug 2020 to Dec 2020	13,80,000	No	Dec 2020	Yes	Yes

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Sanjay P.



10.	NHSRCL	Pre Construction building/ structure condition survey for construction activities of Palghar district of MAHSR project in 46.633 km to 155.718 km for width of 50 m on either side of corridor	May 2019 to March 2021	72,58,526	No	31.03.2021	Yes	Yes
11.	Govt of Odisha	Rapid assessment of past achievement of Biju Setu Yojana	Nov 2021 to April 2022	44,25,000	No	28.04.2022	Yes	
12.	TNRDC-Tamil Nadu Road Development Company Ltd	Appointment of supervision consultant for the work of "Development of Comprehensive Integrated Traffic Infrastructure at TIDEL Park Junction including 'U-shaped flyovers at TIDEL Park junction and Indira Nagar Junction in Rajiv Gandhi Salai (OMR).	Oct 2020 to Nov 2024	1,31,67,000	No	Nov 2024	Yes	Yes

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Maharashtra, Odisha, Tamil Nadu
Telangana, Uttar Pradesh & West Bengal

Sanjay P.



13.	Adani Road Transport Limited	Providing Consultancy services for conducting technical audit for quality of executed works at Azhiyur to Vengalam section of NH-1 (New NH-66) from Des Ch 189+200 (Ex. Km 188+000) to Des Ch.232+100 (Ex Km230+400)[except 6 laning of paloli to Moorad section from Des. Ch. 203+000 (Ex. Km 201+590) to Des. Ch. 205+100 (Ex. Km203+900)] in the State of Kerala Length in KM - 40.8	Sept 2021 to Jan 2025	2,38,12,129	No	Jan 2025	Yes	Yes
14.	Chennai Metro Rail Limited	Independent Safety Audit services of the Signalling & Train control system & its safety related interfaces to rolling stock & platform screen doors systems of phase I extension of CMRL	July 2020 to	81,36,981	Yes		Yes	

Date: 22.01.2026

Place: Mumbai

Signature

(Authorised signatory)

Full Name: Surajit Sankar Pan

Designation: Director, B&I, CIF, South Asia Region

Bureau Veritas (India) Private Limited
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 Telangana, Uttar Pradesh & West Bengal

DTCP HARYANA



To,

30 November 2022

District Town Planner, Enforcement Gurugram,
Department of Town & Country Planning, Haryana;
Sector-14, Huda Complex, 3rd Floor,
Gurugram, Haryana

Ref: Your memo no. GN/DTP-E/2022/9869 dated 12th September 2022

Sub: Work of Structural Safety Audit by conducting detailed visual survey of Multistoried Buildings in Gurugram by empanelled consultants.

Dear Sir,

With aforesaid reference, we, hereby, submit the final Structural Safety Audit reports of all four projects. Details are as below

Sl. No.	Name and Address of Projects assigned	Description
1	Brisk Lumbini, Sector 109	Hard copy of report submitted.
2	Mapsko Royal, Mapsko Paradise, Sector-82	Hard copy of report submitted.
3	Raheja Vedanta Sector-108	Hard copy of report submitted.
4	Antariksh Heights	Hard copy of report submitted.

Yours truly,

For Bureau Veritas India Private Limited.



Durgendra Kumar
B&I Manager, North Region.
BVIL



OFFICE OF DISTRICT TOWN PLANNER (ENFORCEMENT), GURUGRAM
DEPARTMENT OF TOWN & COUNTRY PLANNING, HARYANA
Sector-14, HUDA COMPLEX, 3rd FLOOR, GURUGRAM, Tel-0124-2223749
E-mail: dtpenf3.gurugram.tcp@gmail.com

To

M/s Bureau Vertias,
C-7, Sector 3, 1st Floor,
Noida -201,301(Uttar Pardesh).

Memo No.GN/DTP-E/2022/9869

Dated: 12.09.2022

Subject: Work of Structural Safety Audit by conducting detailed visual survey of Multistoried Buildings in Gurugram by empanelled consultants.

Ref:- In continuation to this office memo no. 7546 dated 05.08.2022.

In reference to your empanelment for the work of Structural Safety Audit of Multistoried Buildings in Gurugram conveyed vide letter under reference followed by the detailed discussions held with you on 04.04.2022 and 31.08.2022, the committee is pleased to accept your rate for the Work of Structural Safety Audit by conducting detailed Visual Survey of Multistoried Buildings in Gurugram at the rates and conditions noted below with a time limit of 45 days to be reckoned from the date of issuance of this letter: -

Sr. No.	Description Of Items	Unit	Rate
1	Conducting detailed visual survey of each building (internally and externally) by visual inspection to record observation of distressed part of building, level of deterioration, damage of slab, columns, beams etc. due to any reason and suggest remedial measures, if any, as regards the issues not endangering the structural safety of building and further recommendation of the type, frequency and location for conducting other tests, if any, as regards to the issues that affect the structural integrity of the building.	sqft	Rs. 1.75 per sft. GST to be paid extra.

Payment Plan :

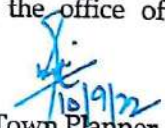
S.No	Stage	Payment to be released within 10 days of completion of stage	Cumulative Payment
1	Release of Work Order	25%	25%
2	Completion of Visual Inspection	25%	50%
3	Submission of detailed report	50%	100%

Assigned Projects :

S.No.	Name and address of project	Site area	Number of towers/ buildings	Tentative total built up area (sft)
1	Antariksh Heights, Sector-84	23.10 Acres	13 Towers	22,89,600.00
2	Brisk Lumbini, BRISK, Sector-109	15.612 Acres	22 Towers	11,12,600.00
3	Mapsko Royal, Mapsko Paradise, Sector-82	17.168 Acres	15 Towers	19,00,000.00
4	Raheja Vedanta Sector-108	10.668 Acres	11 Towers	11,09,650.00
Total tentative area				64,11,850.00

(Note:- The above tentative built up area is subject to the verification of site during inspection)

1. You are hereby requested to furnish a performance security in the form of Bank Guarantee for an amount equivalent to **Rs.6,60,000/- Rupees Six Lakhs Sixty Thousand only** (approximately 5% of the tentative total area) within seven days of the receipt of this letter of acceptance valid upto 31.12.2022. The performance security should be furnished in the favour of District Town Planner, Enforcement, Gurugram which will be released only after expiry of the abovementioned period under written order of the DTP (E), Gurugram.
2. The enquiry committee does not have GSTN. Therefore, you are responsible for depositing GST amount as per rule and furnish the proof of such payment of the committee in subsequent month.
3. The enquiry report to be submitted in the office of District Town Planner, Enforcement, Gurugram.
4. The relevant drawings of the abovementioned projects to be collected from the office of undersigned.


District Town Planner,
Enforcement, Gurugram.

Endst No. GN/DTP-E/2022/ 9870-72

Dated: 12.09.2022

A copy is forwarded to the following for information please:-

1. Deputy Commissioner, Gurugram.
2. Senior Town Planner, Gurugram Circle, Gurugram.
3. Superintending Engineer, PWD (B&R), Gurugram.


District Town Planner,
Enforcement, Gurugram.

Endst No. GN/DTP-E/2022/ 9873

Dated: 12.09.2022

A copy is forwarded to the Director General, Town & Country Planning, Haryana, Chandigarh for information please.


District Town Planner,
Enforcement, Gurugram.

NAVI MUMBAI INTERNATIONAL AIRPORT

NAVI MUMBAI INTERNATIONAL AIRPORT PVT. LIMITED

SERVICE ORDER (SO) AMENDMENT - 02

Consultant: BUREAU VERITAS INDIA PVT. LTD. Ground Floor, Marol Industrial, 72 Business Park, Area Opp. SEEPZ gate, MIDC cross road - C Andheri East Mumbai - 400093, 13 MAHARASHTRA, INDIA. PAN: AABC6767B GSTIN:27AABC6767B1ZS Vendor Code:0000218442 Attention: Mr. Sameer Krishna Rao (Contracts dept.) Mobile No: +91 97691 96559 Email: sameerkrishna.rao@bureauveritas.com	SO, Title	Providing Quality & Environmental Health Safety Inspection Services of Cut & Fill Works at the NMIA Project.
	Agreement No. Dated	5700304681 07 th Feb, 2022
	Amendment No. Amendment dated	02 31 st October, 2023
	Effective Date	01 st November, 2023

This contract Amendment Number 02 to the Original Service Order - 5700304681 is issued to incorporate: (a) change in quantities of existing BoQ items, and (b) extension of service duration.

Details of Amendments:

A. Cost Impact:

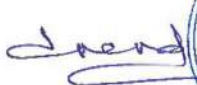
Value of Original Agreement including previous amendments (excl. GST)	Rs. 2,91,72,201 /-
This Amendment (AMD 02) value (excl. GST)	Rs. 49,56,000 /-
Revised Total Contract value (excl. GST)	Rs. 3,41,28,201/-

B. Time Impact:

Completion Schedule as per Original Agreement including previous amendments	31 st Dec., 2024
Completion date as per current Amendment (Amendment 2)	31 st Dec., 2024
Revised Completion Schedule as per this Amendment No. 2	31st Dec., 2024

The price and schedule adjustment (if any) made through this amendment constitute all costs and taxes and includes time adjustment for the full services including the ripple effect associated with the above services. All terms and conditions of the original Contract will apply except for the modification incorporated through this and previous amendments if any to the Original Contract.

For Navi Mumbai International Airport Pvt. Ltd.



Authorized Signatory

Accepted
For BUREAU VERITAS INDIA PVT. LTD.



Authorized Signatory




NAVI MUMBAI INTERNATIONAL AIRPORT PVT. LIMITED

SERVICE ORDER (SO) AMENDMENT – 02

Details of Amendment:

Following clause No. 2 of **Annexure 1.1 – Pricing, Quantity and data** of existing Service order stands cancelled and replaced as shown below:

1. MAN-MONTH RATES

Man-month rates shall be all-inclusive unit prices and shall remain firm & fixed and shall not be subject to change throughout the period of performance under this Contract as per the Conditions of Contract.

Payment according to man-month rates provided under Schedule of Quantity & Rates shall constitute full payment for the provision of personnel shall include but not limited to all cost associated with following:

- Salaries, bonus, notice pay/ retrenchment benefits, provident fund,
- Mobilization and Demobilization
- Replacement personnel in case of rejection of personnel by NMIAL,
- For difficulty of work normally encountered in the projects of this nature, hard working conditions.
- Extra payments to personnel for working on holidays, sundays as & when requisitioned by NMIAL,
- All costs associated with bonus, holidays with pay, sick pay, redundancy,
- Replacement of personnel in place of person leaving/resigned, absent, proceeding on leave, person not attended after shift duty hours,
- Transportation costs of personnel commuting from place of work to place of stay & otherwise.
- Statutory employee benefits which Service Provider makes either directly to the employees or deposits with the State or any legal trusts under National or State Laws, & full compliance with all statutory Government regulations & applicable law, and
- Working in shifts if required.
- All taxes and costs of whatever nature except GST.
- Head office and site office overheads, providing Authorized Representative, Time Keeping, Accounting, Clerical support, Labour welfare, & Profits.

No payments shall be made for period during which personnel are not requisitioned for working on the site, Force Majeure days, Suspension by NMIAL, Lunch time or for off Working Hours, absence, leave or vacation time.

The quantities and Estimated Contract Price mentioned below are not minimum guaranteed quantities and/or prices.

2. Quantities

Service Provider shall mobilize (and demobilize) personnel as directed by NMIAL from time to time. The quantities applicable to this Contract will be based on the NMIAL requirement, actual number of personnel so mobilized (or demobilized), and the number of man months.

The Estimated Revised Contract Price is Rs. 3,41,28,201/- (Rupees Three Hundred Forty-One Lakhs Twenty-eight Thousand Two Hundred one only). Excluding Goods and Services Tax (GST).



Sanjay P.

NAVI MUMBAI INTERNATIONAL AIRPORT PVT. LIMITED

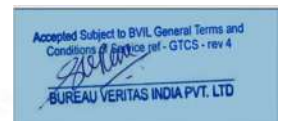
SERVICE ORDER (SO) AMENDMENT – 02

The Estimated Contract Price as mentioned above is estimated on man-month rate mentioned in Table A2- Schedule of Quantity & man-month rates are valid up to 31st Dec 2024 and are based on scope mentioned in Annexure 2 – Scope of Services. The deployed QA/QC and EHS staff from the service provider must perform his/her duty in the entire NMIA project premise as, where and when requires including but not limited to EPC-1&2 contract, non-EPC buildings/structures, other Misc. structures, boundary wall etc. Table-A and Table-A1 issued earlier via. amendment-1 shall remain valid till revised contract completion date mentioned in this amendment-2. The man-month rate includes any and all direct and indirect costs of whatsoever nature, including all profit, royalty and other fees, the cost of all manpower resources, materials, equipment's, software, special tools, stationary, Taxes (including all deductions and withholdings but excluding Goods and Service Tax (GST)), mobilization, demobilization, insurance premiums, travel, visa (if any), boarding-lodging, telecommunication, courier & administrative expenses, license fees relating to or arising out of the performance of the Services under the Agreement and the achievement of completion of Services. GST as applicable shall be paid extra.

The man-month rates shall be firm and fixed all-inclusive price for carrying out the Services and is not subject to escalation on any grounds whatsoever.

Table A - Schedule of Quantity & Man-month Rates (Applicable till 31.08.2023)

S. No.	DESCRIPTION	QTY	Uom	RATE (INR)	AMOUNT (INR)
	SAC No-				
1	Providing of Quality (QA/QC) Inspection Engineer (Civil) with qualification and experience as mentioned under Scope of Services, clause No. 3.1.1 as on need basis Experience (Minimum 10 yrs Diploma or 6 yrs Degree in Civil)- 04 resources for 16 months	68.833	Man moths	85,000	5,850,771
2	Providing of Environmental Health Safety (EHS) Executives with qualification and experience as mentioned under Scope of Services, clause No. 3.2.1 as on need basis (Minimum 3 yrs exp with Degree/diploma and safety management qualifications as mentioned in scope)- 06 resources for 16 months	104.958	Man moths	85,000	8,921,430
Estimated Revised Contract Price Rs. (A)					14,772,201.00



NAVI MUMBAI INTERNATIONAL AIRPORT PVT. LIMITED

SERVICE ORDER (SO) AMENDMENT – 02

Table A1 - Schedule of Quantity & Man-month Rates (Applicable from 01.09.2023 to 31.12.2024)

S. No.	DESCRIPTION	QTY	Uom	RATE (INR)	AMOUNT (INR)
1	Providing of Quality (QA/QC) Inspection Engineer (Civil) with qualification and experience as mentioned under Scope of Services, clause No. 3.1.1 as on need basis Experience (Minimum 10 yrs Diploma or 6 yrs Degree in Civil)- 04 resources for 16 months	64	Man moths	90,000	5,760,000
2	Providing of Environmental Health Safety (EHS) Executives with qualification and experience as mentioned under Scope of Services, clause No. 3.2.1 as on need basis (Minimum 3 yrs exp with Degree/diploma and safety management qualifications as mentioned in scope)- 06 resources for 16 months	96	Man moths	90,000	8,640,000
Estimated Revised Contract Price Rs. (B)					14,400,000.00

Table A2 for Variation work - Schedule of Quantity & Man-month Rates (Applicable from 01.11.2023 to 31.12.2024)

S. No.	DESCRIPTION	QTY	Uom	RATE (INR)	AMOUNT (INR)
1	Providing of Quality (QA/QC) Inspection Engineer (Mech-Structural steel)) with qualification and experience as mentioned under Scope of Services, clause No. 3.1.1 as on need basis Experience (10 yrs. Diploma or 7 yrs. Degree)	14	Man moths	118,000	1,652,000
2	Providing of Quality (QA/QC) Inspection Engineer (MEP-Mechanical)) with qualification and experience as mentioned under Scope of Services, clause No. 3.1.1 as on need basis Experience (10 yrs. Diploma or 7 yrs. Degree)	14	Man moths	118,000	1,652,000
3	Providing of Quality (QA/QC) Inspection Engineer (MEP-Electrical)) with qualification and experience as mentioned under Scope of Services, clause No. 3.1.1 as on need basis Experience (10 yrs. Diploma or 7 yrs. Degree)	14	Man moths	118,000	1,652,000
Estimated Revised Contract Price Rs. (C)					4,956,000.00
Revised Contract Value Rs. (A) + (B) + (C)					34,128,201.00
GST @ 18 %					6,143,076.18
Total Rs.					40,271,277.18
NOTE	For Table A, A1 & A2				
	Above stated quantities are estimates and are not minimum guaranteed quantities.				
	Service Provider may be asked to increase or decrease no. of engineers depending upon NMIAL's requirements with prior intimation of 10 days.				
	Deployment of staff mentioned in Table A2 shall be exclusively with prior approval of NMIAL and only after issuance of notice to proceed.				



NAVI MUMBAI INTERNATIONAL AIRPORT PVT. LIMITED

SERVICE ORDER (SO) AMENDMENT - 02

3. MEASUREMENT FOR PAYMENT

Service Provider shall submit monthly invoices (RA Bills) in accordance with the man month quantities certified by the authorized representative of NMIAL & man-month rates against duly certified quantities. Within ten (10) calendar days after invoice receipt, NMIAL may reject all or part of such invoice stating the deficiencies, if any, and advising the Service Provider for correction and compliance in accordance with the provisions of the contract.

Invoices shall be made out to Navi Mumbai International Airport Pvt. Ltd. on the following Address:
Navi Mumbai International Airport Pvt Ltd,
1st Floor, Office of Airport Director, Terminal 1B,
CSI Airport, Santacruz (E), Mumbai - 400 099, Maharashtra, India

and delivered in 2 originals on the following address:
Navi Mumbai International Airport Pvt. Ltd. (NMIAL)
V Times Square, 11th Floor, Plot No. 3, Sector 15, CBD Belapur, Navi Mumbai 400 614
Attention: Mr. Ravindra Dalvi

Reference: SO No: 5700304681

Upon submission of commercially correct invoice, NMIAL shall release the payments within 30 (thirty) calendar days from the date of submission of the correct invoice.

Payment for above items shall be made based on actual deployment certified by NMIAL at a man month rate specified above in Table A Schedule of Quantity & Rates. In the event of performance

of duties for part of the month, pro-rate payment based on number of days worked divided by number of days in the particular month shall be made.

Man month's rate shall be paid only in accordance with daily attendance muster/ register/ time sheets approved by NMIAL. Copy of the approved register shall be attached to the request for payment.

4. Evidence of Payment:

Service Provider shall submit the evidence of payments made to its staff in terms of pay slips, signed wages register etc. required by Law, whenever requested by NMIAL. Service Provider shall handover copies of relevant documents and records maintained in connection with statutory details/ deposits. Maintain records at the Service Provider offices for reconciliation as and when required by law. Service Provider shall produce the original stamped PF challans receipts and other necessary documents for verification if requested by NMIAL.

5. RECORDS:

A Time Sheet/ daily attendance muster/ register for each personnel in format acceptable to NMIAL shall be implemented. Approved Time Sheet/ daily attendance muster/ register shall be attached with every months invoice for verification.

All records of the Contract arising out of Statutory Requirement shall be maintained by the Service Provider for a minimum period of 3 years after demobilization, or more when required by law or statutory requirements. They shall be available to the NMIAL as and when directed to be provided.

6. GOODS AND SERVICES TAX ("GST"):

All Taxes shall be to the account of and borne by the Service Provider and shall be paid directly by the Service Provider to the government authority concerned. Service Provider shall be responsible for all procedural compliances related to payment of Taxes and shall solely be responsible for any proceedings initiated by the relevant government authority, inter alia, in respect of any non-payment, short-payment, non-compliance, penalty, interest or other such issue, and for all liabilities and expenses related to such proceedings.

The NMIAL shall be entitled to deduct/ withhold in accordance with applicable law, income tax or withholding tax, or other deductions (as the case may be) from any payments made to the Service Provider and the amount so deducted shall be deemed to be a payment made to



NAVI MUMBAI INTERNATIONAL AIRPORT PVT. LIMITED

SERVICE ORDER (SO) AMENDMENT – 02

the Service Provider. The NMIAL shall provide the appropriate tax certificates to the Service Provider evidencing the Taxes deducted at source.

GOODS AND SERVICES TAX (GST):

- (a) in this clause unless expressly stated otherwise in the Service Order, words and expressions which have a defined meaning in the GST Act have the same meaning as in this Service Order;
- (b) Notwithstanding the definition of consideration in the GST Act, and unless expressly stated otherwise in the Service Order, prices or other sums payable or consideration to be provided under or in accordance with the Service Order are exclusive of GST.
- (c) All supply or Services provided shall be made with 'tax invoice' in compliance with Rules and Regulations of GST law by clearly stating the codes of HSN/ SAC as applicable along with the specified taxes as per the Service Order.
- (d) Service Provider shall make GST payment to Government treasury and file the requisite periodical GST outward returns as per law enabling NMIAL to avail the GST credit. NMIAL reserves the right (i) to recover / withhold the value equivalent to double the amount of GST for which compliances are pending and other damages resulting from noncompliance under GST law from the amounts payable to Service Provider, if Service Provider fails to comply with the GST regulations including but not limited to the payment of Taxes and filing of the correct return within the time period as may be notified. Such withholding shall be against the Service Provider's invoice amount under consideration and the Service Provider shall file the returns as per the invoice raised. (ii) to terminate the order / contract in case tax credit issues are persisting for three months consecutively including that of the GST merit rating of the Service Provider. Rectification of mismatches shall be done at the Service Provider's end within the timelines as prescribed under the GST laws. Any loss of eligible input credit and consequential interest/ penalty costs to NMIAL arising out of such mismatch shall be made good by the Service Provider.
- (e) In the event an adjustment arises in connection with a supply made or services provided under this Service Order, the Service Provider must give the other party a credit note or debit note in accordance with the GST Act and regulations made thereunder.
- (f) If this Service Order requires one party to pay for, reimburse or contribute to any expense, loss or outgoing suffered or incurred by the other party, the amount required to be paid, reimbursed or contributed by the first party will be reduced by the amount of input tax credits (if any) to which the other party is entitled in respect of the reimbursable expense.
- (g) All payments under the Service Order shall be subject to TDS/TCS rules notified under the tax laws including GST law prevailing in India.
- (h) Service Provider hereby agrees to address all the queries raised by NMIAL and shall amend, rectify and/or revise the return in case of error/ omissions etc. and comply with GST law requirements, failing which NMIAL reserves the right to take suitable action including foreclosure of the Service Order.

All other Terms and Conditions of Service Order amendment-1 shall remain unchanged.



Signature
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